

Statistics By Jim

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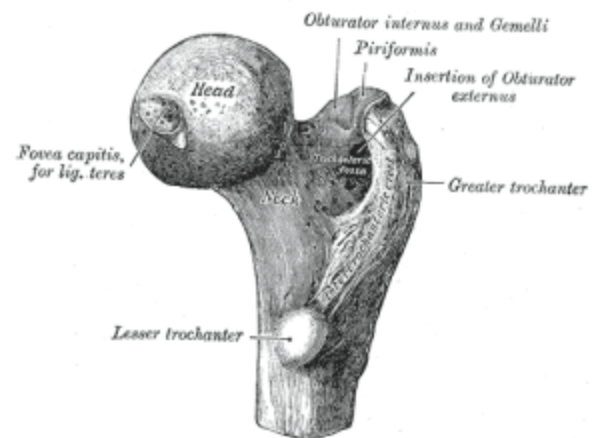
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Multicollinearity in Regression Analysis: Problems, Detection, and Solutions

By [Jim Frost](#) — [226 Comments](#)

Multicollinearity occurs when independent variables in a regression model are correlated. This correlation is a problem because independent variables should be *independent*. If the degree of correlation between variables is high enough, it can cause problems when you fit the model and interpret the results.

In this blog post, I'll highlight the problems that multicollinearity can cause, show you how to test your model for it, and highlight some ways to resolve it. In some cases, multicollinearity isn't necessarily a problem, and I'll show you how to make this determination. I'll work through an example dataset which contains multicollinearity to bring it all to life!



I use regression to model the bone mineral density of the femoral neck in order to, pardon the pun, flesh out the effects of multicollinearity. Image By Henry Vandyke Carter – Henry Gray (1918)



07:17

Why is Multicollinearity a Potential Problem?

A key goal of regression analysis is to isolate the relationship between each independent variable and the dependent variable. The interpretation of a regression coefficient is that it represents the mean change in the dependent variable for each 1 unit change in an independent variable when you *hold all of the other independent variables constant*. That last portion is crucial for our discussion about multicollinearity.

The idea is that you can change the value of one independent variable and not the others. However, when independent variables are correlated, it indicates that changes in one variable are associated with shifts in another variable. The stronger the correlation, the more difficult it is to change one variable without changing another. It becomes difficult for the model to estimate the relationship between each independent variable and the dependent variable *independently* because the independent variables tend to change in unison.

There are two basic kinds of multicollinearity:

- **Structural multicollinearity:** This type occurs when we create a model term using other terms. In other words, it's a byproduct of the model that we specify rather than being present in the data itself. For example, if you square term X to model curvature, clearly there is a correlation between X and X^2 .

- **Data multicollinearity:** This type of multicollinearity is present in the data itself rather than being an artifact of our model. Observational experiments are more likely to exhibit this kind of multicollinearity.

Related post: [What are Independent and Dependent Variables?](#)

What Problems Do Multicollinearity Cause?

Multicollinearity causes the following two basic types of problems:

- The coefficient estimates can swing wildly based on which other independent variables are in the model. The coefficients become very sensitive to small changes in the model.
- Multicollinearity reduces the precision of the estimated coefficients, which weakens the statistical power of your regression model. You might not be able to trust the p-values to identify independent variables that are statistically significant.

Imagine you fit a regression model and the coefficient values, and even the signs, change dramatically depending on the specific variables that you include in the model. It's a disconcerting feeling when slightly different models lead to very different conclusions. You don't feel like you know the actual effect of each variable!

Now, throw in the fact that you can't necessarily trust the p-values to select the independent variables to include in the model. This problem makes it difficult both to [specify the correct model](#) and to justify the model if many of your p-values are not statistically significant.

As the severity of the multicollinearity increases so do these problematic effects. However, these issues affect only those independent variables that are correlated. You can have a model with severe multicollinearity and yet some variables in the model can be completely unaffected.

The regression example with multicollinearity that I work through later on illustrates these problems in action.

Do I Have to Fix Multicollinearity?

Multicollinearity makes it hard to interpret your coefficients, and it reduces the power of your model to identify independent variables that are statistically significant. These are definitely

serious problems. However, the good news is that you don't always have to find a way to fix multicollinearity.

The need to reduce multicollinearity depends on its severity and your primary goal for your regression model. Keep the following three points in mind:

1. The severity of the problems increases with the degree of the multicollinearity. Therefore, if you have only moderate multicollinearity, you may not need to resolve it.
2. Multicollinearity affects only the specific independent variables that are correlated. Therefore, if multicollinearity is not present for the independent variables that you are particularly interested in, you may not need to resolve it. Suppose your model contains the experimental variables of interest and some control variables. If high multicollinearity exists for the control variables but not the experimental variables, then you can interpret the experimental variables without problems.
3. Multicollinearity affects [the coefficients and p-values](#), but it does not influence the predictions, precision of the predictions, and the goodness-of-fit statistics. If your primary goal is to make predictions, and you don't need to understand the role of each independent variable, you don't need to reduce severe multicollinearity.

Over the years, I've found that many people are incredulous over the third point, so here's a reference!

“ The fact that some or all predictor variables are correlated among themselves does not, in general, inhibit our ability to obtain a good fit nor does it tend to affect inferences about mean responses or predictions of new observations. —Applied Linear Statistical Models, p289, 4th Edition.

If you're performing a designed experiment, it is likely orthogonal, meaning it has zero multicollinearity. Learn more about [orthogonality](#).

Testing for Multicollinearity with Variance Inflation Factors (VIF)

If you can identify which variables are affected by multicollinearity and the strength of the correlation, you're well on your way to determining whether you need to fix it. Fortunately, there is a very simple test to assess multicollinearity in your regression model. The variance inflation

factor (VIF) identifies correlation between independent variables and the strength of that correlation.

Statistical software calculates a VIF for each independent variable. VIFs start at 1 and have no upper limit. A value of 1 indicates that there is no correlation between this independent variable and any others. VIFs between 1 and 5 suggest that there is a moderate correlation, but it is not severe enough to warrant corrective measures. VIFs greater than 5 represent critical levels of multicollinearity where the coefficients are poorly estimated, and the p-values are questionable.

Use VIFs to identify correlations between variables and determine the strength of the relationships. Most statistical software can display VIFs for you. Assessing VIFs is particularly important for observational studies because these studies are more prone to having multicollinearity.

Multicollinearity Example: Predicting Bone Density in the Femur

This regression example uses a subset of variables that I collected for an experiment. In this example, I'll show you how to detect multicollinearity as well as illustrate its effects. I'll also show you how to remove structural multicollinearity. You can download the CSV data file:

[MulticollinearityExample](#).

I'll use regression analysis to model the relationship between the independent variables (physical activity, body fat percentage, weight, and the interaction between weight and body fat) and the dependent variable (bone mineral density of the femoral neck).

Here are the regression results:

Regression Analysis: Femoral Neck versus %Fat, Weight kg, Activity

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	0.555785	0.138946	27.95	0.000
%Fat	1	0.009240	0.009240	1.86	0.176
Weight kg	1	0.127942	0.127942	25.73	0.000
Activity	1	0.047027	0.047027	9.46	0.003
%Fat*Weight kg	1	0.041745	0.041745	8.40	0.005
Error	87	0.432557	0.004972		
Total	91	0.988342			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.0705118	56.23%	54.22%	50.48%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0.155	0.132	1.18	0.243	
%Fat	0.00557	0.00409	1.36	0.176	14.93
Weight kg	0.01447	0.00285	5.07	0.000	33.95
Activity	0.000022	0.000007	3.08	0.003	1.05
%Fat*Weight kg	-0.000214	0.000074	-2.90	0.005	75.06

These results show that Weight, Activity, and the interaction between them are statistically significant. The percent body fat is not statistically significant. However, the VIFs indicate that our model has severe multicollinearity for some of the independent variables.

Notice that Activity has a VIF near 1, which shows that multicollinearity does not affect it and we can trust this coefficient and p-value with no further action. However, the coefficients and p-values for the other terms are suspect!

Additionally, at least some of the multicollinearity in our model is the structural type. We've included the interaction term of body fat * weight. Clearly, there is a correlation between the interaction term and both of the main effect terms. The VIFs reflect these relationships.

I have a neat trick to show you. There's a method to remove this type of structural multicollinearity quickly and easily!

Center the Independent Variables to Reduce Structural Multicollinearity

In our model, the interaction term is at least partially responsible for the high VIFs. Both higher-order terms and interaction terms produce multicollinearity because these terms include the main effects. Centering the variables is a simple way to reduce structural multicollinearity.

Centering the variables is also known as standardizing the variables by subtracting the mean. This process involves calculating the mean for each continuous independent variable and then subtracting the mean from all observed values of that variable. Then, use these centered variables in your model. Most statistical software provides the feature of fitting your model using standardized variables.

There are other standardization methods, but the advantage of just subtracting the mean is that the interpretation of the coefficients remains the same. The coefficients continue to represent the mean change in the dependent variable given a 1 unit change in the independent variable.

In the worksheet, I've included the centered independent variables in the columns with an S added to the variable names.

For more about this, read my post about [standardizing your continuous independent variables](#).

Regression with Centered Variables

Let's fit the same model but using the centered independent variables.

Regression Analysis: Femoral Neck versus %Fat S, Weight S, Activity S

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	0.55578	0.138946	27.95	0.000
%Fat S	1	0.04786	0.047863	9.63	0.003
Weight S	1	0.30473	0.304728	61.29	0.000
Activity S	1	0.04703	0.047027	9.46	0.003
%Fat S*Weight S	1	0.04175	0.041745	8.40	0.005
Error	87	0.43256	0.004972		
Total	91	0.98834			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.0705118	56.23%	54.22%	50.48%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0.82161	0.00973	84.40	0.000	
%Fat S	-0.00598	0.00193	-3.10	0.003	3.32
Weight S	0.00835	0.00107	7.83	0.000	4.75
Activity S	0.000022	0.000007	3.08	0.003	1.05
%Fat S*Weight S	-0.000214	0.000074	-2.90	0.005	1.99

The most apparent difference is that the VIFs are all down to satisfactory values; they're all less than 5. By removing the structural multicollinearity, we can see that there is some multicollinearity in our data, but it is not severe enough to warrant further corrective measures.

Removing the structural multicollinearity produced other notable differences in the output that we'll investigate.

Comparing Regression Models to Reveal Multicollinearity Effects

We can compare two versions of the same model, one with high multicollinearity and one without it. This comparison highlights its effects.

The first independent variable we'll look at is Activity. This variable was the only one to have almost no multicollinearity in the first model. Compare the Activity coefficients and p-values between the two models and you'll see that they are the same (coefficient = 0.000022, p-value = 0.003). This illustrates how only the variables that are highly correlated are affected by its problems.

Let's look at the variables that had high VIFs in the first model. The standard error of the coefficient measures the precision of the estimates. Lower values indicate more precise estimates. The standard errors in the second model are lower for both %Fat and Weight. Additionally, %Fat is significant in the second model even though it wasn't in the first model. Not only that, but the coefficient sign for %Fat has changed from positive to negative!

The lower precision, switched signs, and a lack of statistical significance are typical problems associated with multicollinearity.

Now, take a look at the Summary of Model tables for both models. You'll notice that the **standard error of the regression (S)**, **R-squared**, **adjusted R-squared**, and **predicted R-squared** are all identical. As I mentioned earlier, multicollinearity doesn't affect the predictions or goodness-of-fit. If you just want to make predictions, the model with severe multicollinearity is just as good!

How to Deal with Multicollinearity

I showed how there are a variety of situations where you don't need to deal with it. The multicollinearity might not be severe, it might not affect the variables you're most interested in, or maybe you just need to make predictions. Or, perhaps it's just structural multicollinearity that you can get rid of by centering the variables.

But, what if you have severe multicollinearity in your data and you find that you must deal with it? What do you do then? Unfortunately, this situation can be difficult to resolve. There are a variety of methods that you can try, but each one has some drawbacks. You'll need to use your subject-area knowledge and factor in the goals of your study to pick the solution that provides the best mix of advantages and disadvantages.

The potential solutions include the following:

- Remove some of the highly correlated independent variables.
- Linearly combine the independent variables, such as adding them together.
- Partial least squares regression uses **principal component analysis** to create a set of uncorrelated components to include in the model.
- LASSO and Ridge regression are advanced forms of regression analysis that can handle multicollinearity. If you know how to perform linear least squares regression, you'll be able to handle these analyses with just a little additional study.

As you consider a solution, remember that all of these have downsides. If you can accept less precise coefficients, or a regression model with a high R-squared but hardly any statistically significant variables, then not doing anything about the multicollinearity might be the best solution.

In this post, I use VIFs to check for multicollinearity. For a more in-depth look at this measure, read my post about [Calculating and Assessing Variance Inflation Factors \(VIFs\)](#).

If you're learning regression and like the approach I use in my blog, check out my [Intuitive Guide to Regression Analysis](#) book! You can find it on Amazon and other retailers.

Comments

[hendrickmaulanaibrahim](#) says
May 15, 2025 at 9:53 pm

Can you explain the difference between perfect multicollinearity and high multicollinearity?

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Reply

Atikhom says

April 2, 2025 at 9:54 am

I'm using multinomial logistic regression. I've found that two variables are highly correlated (VIF = 30.837 and 49.401. Since both variables are of theoretical importance, it is recommended by a reviewer that regularization would strengthen the methodology. I would appreciate it if you could provide step by step advice on how to perform regularization.

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Reply

Richard H. Serlin says

April 1, 2025 at 11:46 pm

I'm an old man now, and it's been a very long time since grad school, but this came up recently looking at a regression of foods on healthy aging. These foods are highly intercorrelated; for example, the amount of whole grains a person eats is very negatively correlated with the amount of refined grains: eat more whole wheat bread, and you eat less white bread. The study, however, has a ton of data, 105,000 doctors and nurses followed for 30 years. The results are very interesting:

<https://erictopol.substack.com/p/our-diet-and-healthy-aging>

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Reply

Mekuriaw says

January 21, 2025 at 2:42 pm

Thank you dear!

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[Reply](#)

Ricky Jordan says

September 29, 2024 at 12:33 am

With regard to the statement “If your primary goal is to make predictions, and you don’t need to understand the role of each independent variable, you don’t need to reduce severe multicollinearity” – can you please give an example of this? Thank you.

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[Reply](#)

Jim Frost says

September 29, 2024 at 11:12 pm

Hi Ricky,

Sure! In this post, I present a model that uses physical activity, body fat percentage, weight, and the interaction between weight and body fat to predict the femoral neck bone density. However, this model has multicollinearity. The multicollinearity makes it difficult to trust and interpret the coefficients and p-values. We can’t be sure whether the variables are significant nor whether their coefficients are accurate. Despite these problems, if we enter values into the regression equation for the IVs, we’ll get a valid [prediction](#) (assuming we satisfy the [OLS assumptions](#)).

In other words, multicollinearity affects the interpretability of the IVs but not the model’s predictions.

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[Reply](#)

Ronrico A Mangapit says

[September 27, 2024 at 4:41 am](#)

Jim, as for the statement above – “If your primary goal is to make predictions, and you don’t need to understand the role of each independent variable, you don’t need to reduce severe multicollinearity” – can you please give an example for this? Thank you.

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[Reply](#)

Kyle says

[September 26, 2024 at 1:36 am](#)

Hi Jim,

I have been involved in various studies where the aim is to predict tumour uptake of a therapeutic agent or diagnostic tracer using variables like disease or patient characteristics.

For instance, we might have tumour uptake (continuous) as the dependent variable, and tumour size (continuous) and age (continuous) as the independent variables.

Researchers, especially those publishing, are often familiar with linear regression. However, a patient can have multiple tumours, each with its own uptake value. This violates the assumption of independent observations in linear regression and also inflates the sample size.

The imaging specialists in my company make a point that the correlation between tumour type and size for example is more evident than the correlation between tumours within the same patient.

My question is, is it appropriate to perform linear regression in such cases, disregarding the multiple tumours per subject and focusing on a per-tumour analysis? Or should I strongly recommend an alternative approach like linear mixed models, which can account for the multiple tumours within a patient?

Thanks!

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[Reply](#)

Jim Frost says

September 26, 2024 at 10:25 pm

Hi Kyle,

I'm far from an expert in your field. But it comes down to whether you can consider the tumors on the one patient as independent observations. Does the uptake of one tumor on a patient provide information about the other tumors. If the answer is yes, then they're not independent observations and you should use a different methodology, such as a mixed-effects model or generalized estimating equations (GEE). These methods can handle the clustering.

However, if a tumor on a patient does not provide information about another tumor on the same patient, you wouldn't need to use those methods.

It sounds like a correlation is known or suspected for the tumors within a patient. If so, I'd use one of the indicated methods. Alternatively, you could just use one tumor from a patient.

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[Reply](#)

labetterworld says

August 22, 2024 at 11:38 am

really awesome i just read and observe thanks for the astonishing information from labetterworld
Thanks

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[Reply](#)

Sophie says
[June 21, 2024 at 5:42 am](#)

Hi Jim,

I have a question regarding multicollinearity. Is it possible for multicollinearity between two independent variables (the variable of interest and a control variable) to cause the variable of interest to appear significant? I came across a paper suggesting that a previously identified significant relationship between two variables was due to multicollinearity. This seems counterintuitive, as multicollinearity typically leads to insignificance due to inflated standard errors.

Could you provide an explanation for this phenomenon or suggest resources where I can learn more about it?

Thank you in advance!

Best regards
Sophie

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[Reply](#)

Jim Frost says
[June 21, 2024 at 2:53 pm](#)

Hi Sophie,

That's definitely not the typical effect of multicollinearity. As you mentioned, multicollinearity tends to lead to insignificant results. In those cases, removing the multicollinearity can restore the significance. I've never heard of a case where multicollinearity produced significant results and it doesn't make theoretical sense.

Hence, I don't have an explanation for the study's results. Perhaps there is something else also occurring that is confusing the issue?

For example, perhaps the the relationship between the variable of interest and outcome was strong enough so that even with the multicollinearity, it was still significant. Then, in an effort to deal with the multicollinearity, they removed the control variable. In that scenario, it's possible that **omitted variable bias** is now causing the variable of interest to incorrectly appear insignificant. If this were the case, it would suggest that the significant result in the first model was correct and that the later incorrect results are due to something other than multicollinearity.

That's just one possibility out of many. It really depends on a multitude of specific details. I don't know what they did to deal with the multicollinearity to make that claim. But I would suspect there is something else happening.

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[Reply](#)

Oh Jianyong says

September 21, 2024 at 11:34 pm

Not the original commenter but isn't it possible to detect spurious correlations that don't make theoretical sense.

Suppose IV1 (e.g., "years of education") is correlated with IV2 (e.g., "income level"), but only IV2 truly affects the target variable (e.g., "job performance"). In a regression model :

If both IV1 and IV2 are included, the model might estimate a significant coefficient for IV1 due to its correlation with IV2, even if IV1 does not directly contribute to job performance.

This could mislead decision-makers into thinking that increasing years of education directly improves job performance, while the effect is actually driven by income level.

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[Reply](#)

Jim Frost says

[September 26, 2024 at 10:48 pm](#)

Hi,

Yes, multicollinearity cause problems like that, which is a correlation between predictors. This is not a matter of spurious correlations. You need to use your subject-area knowledge of the pattern of correlations in the subject matter to understand what is happening. And, yes, as you say, you need to determine whether the correlations make theoretical sense.

I'm not sure that your final paragraph is factually correct or not. But you could use a method like Ridge Regression to reduce the effects of multicollinearity.

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[Reply](#)

Maria Konstantina Filippakopoulou says

[June 16, 2024 at 10:04 am](#)

Hello! Thank you so much for the useful blog. I hope you are still answering to comments 😊

I am using the Behavioral Reasoning Theory (BRT) to investigate the adoption intention of cryptocurrencies in the Netherlands. I am using SMARTPLS4 to run a consistent PLS SEM, and I have found that there is a lot of multicollinearity in my variables.

For example, the VIF of “attitude towards crypto” → “adoption intention of crypto” is 13! Additionally the VIF of “reasons against adoption of crypto” → “adoption intention of crypto” is 10.7!

Is it normal for there to be multicollinearity when using the Behavioral Reasoning Theory model? If not, how should I proceed, so that I can still make some conclusions and be able to interpret my path coefficients?

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[Reply](#)

Jim Frost says

June 16, 2024 at 8:04 pm

Hi Maria,

Those are extremely high VIFs! I’m not familiar with the Behavioral Reasoning Theory model. I don’t know how normal that is. I would recommend researching that and seeing how they handle it in that field if it is common. Perhaps they have standard methods?

From a statistical standpoint, I’d recommend using LASSO or Ridge regression which are more robust against multicollinearity than regular least squares regression. With that much multicollinearity you don’t want to use ordinary least squares.

I see you’re using partial least squares regression which reduces the data down to uncorrelated components and then performs regression using those components.

That's a good approach for when you have high levels of multicollinearity. Although, it's more difficult to understand the relationships between the variables and the outcome using that method.

I hope that helps!

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[Reply](#)

Sourav says

June 12, 2024 at 12:24 pm

Hi, thank you so much for the post. I am looking into impact of physical infrastructure on GDP, and capital and population are my other explanatory variables. they demonstrated extreme multicollinearity so I tried Partial regression and ridge regression. However, the regression residual shows presence of serial autocorrelation? is it still okay to consider Partial regression and ridge regression in my case?

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[Reply](#)

Jim Frost says

June 12, 2024 at 8:43 pm

Hi Sourav,

Serial autocorrelation, where error terms in a time series are correlated with each other affect the validity of your statistical tests. This is something that neither partial regression nor ridge regression are designed to address directly. These techniques focus on reducing the impact of multicollinearity but do not correct for autocorrelation.

Try including lagged variables to fix the serial correlation. After addressing the serial correlation, you can see if multicollinearity is a problem.

I hope that helps!

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[Reply](#)

Isabella Smith says

May 7, 2024 at 10:29 pm

Hi! Thanks so much for this helpful blog post. Quick question – I’m currently running GLMs and I’m wondering if it’s a problem to include an independent variable if it is correlated with the dependent variable, or is multicollinearity only an issue between independent variable?

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[Reply](#)

Jim Frost says

May 7, 2024 at 11:17 pm

Hi Isabella,

Multicollinearity and its problems only apply to correlations between independent variables (IVs) and does not involve the dependent variable (DV) at all.

In fact, you want to find IVs that correlate with the DV because that means there is a relationship between them that the regression model can find and describe. So, it’s actually a good thing to have a correlation between an IV and the DV!

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[Reply](#)

Jamie Dekkers says

February 12, 2024 at 4:42 am

Als ik twee onafhankelijke variabele heb in mijn regressiemodel, die beide dezelfde vif en tolerance waarden hebben, moet ik ze dan in mijn verslag allebei noemen voor de multicollineariteit?

If I have two independent variables in my regression model, both of which have the same vif and tolerance values, should I mention them both in my report for multicollinearity?

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[Reply](#)

Jim Frost says

February 12, 2024 at 6:15 pm

Hi Jamie,

I translated your original question from Dutch to English.

Yes, at the very least, you should report the VIFs in your tabular results. You can mention that multicollinearity is or isn't a problem in your results as you see fit. Particularly important to mention VIFs that are too high. But if they're ok, you can just include them in a table. I also recommend seeing how others in your field report their results. Some fields have their own standards for what and how to report statistics.

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Reply

Larry A Lutsky says

December 12, 2023 at 6:37 pm

You mention centering as a way of reducing collinearity but do not recommend standardization because this changes the coefficients. Do you recommend standardizing variables before creating interaction terms or just centering? My fear is that if the standard deviations are very different then the variable with the larger standard deviation will overwhelm the one with the smaller standard deviation in the interaction term. Standardization seems to put them on equal footing, so a one unit change on either has the same relative impact and I think this makes sense. Am I right?

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Reply

Jim Frost says

December 12, 2023 at 7:03 pm

Typically, I recommend centering. That's if you want to interpret the coefficients in the usual fashion.

I recommend standardization when you're trying to understand the relative importance of the various predictors. That's the context for when placing them on equal footing is important. However, it's important to note that standardizing changes the interpretation. It's no longer "one unit of change," but rather the coefficients now represent the mean change in the DV given a *one standard deviation change* in the IV. To learn more about this use for standardization, read my post about [Identifying the Most Important Variables in Your Model](#).

It's easier to understand natural effect sizes (centered) intuitively while standardized effect sizes are better for comparing. Both centering and standardization are equally effective for reducing structural multicollinearity. So, it

really depends on what you're trying to learn. In fact, you might even fit the model both ways!

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[Reply](#)

Thu Nguyen says

September 22, 2023 at 1:24 am

Hi sir, i have one question. My data has 2 dummy variables and the VIF result of reg is quite high (24; 22) but has pwcorr below 0.8 and both variables are statistically significant. So what should I do to reduce it or how can I argue in my research paper so that I can continue to use it.

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[Reply](#)

Jim Frost says

September 22, 2023 at 1:37 am

Hi,

Unfortunately, that is very high multicollinearity. Trust the VIF over the pairwise correlations because multiple IVs can contribute to the multicollinearity of another. VIFs can pick that up but pairwise correlations cannot.

I suggest some possibilities in this post. You can see if removing any of the offending variables helps without reducing the goodness of fit by much. Because they are highly correlated, they're largely explaining the same thing. Consequently, sometimes you can remove some of them without reducing the explanatory power of your model by much.

If that doesn't work or otherwise not an option, consider using Ridge or LASSO regression. Those can handle multicollinearity much better than least squares. You can leave in all your IVs.

If you're just using the model to make predictions, then you don't need to worry about it.

Alternatively, if you want to understand the relationships between the variables and the p-values are significant, you might be able to live with the inflated variability. You can note in your paper that the IVs are significant but that the precision of the coefficient estimate is reduced by the multicollinearity. Sometimes that lowered precision makes the variable non-significant. But if yours are significant and you can live with the reduced precision of the estimate, you don't need to do anything. Basically, this means that, yes, the IVs are significant, but their CIs are wider than they would be without the multicollinearity (that's the lower precision of the estimate).

Those are some of your options. I hope that helps!

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[Reply](#)

pourya says

August 16, 2023 at 12:45 pm

Hi Jim!

Thank you for the article, it is very clear and interesting.

I hope you are still answering comments.

Is it correct to say that at the end the only thing that matters is the confidence interval (CI) of the coefficients? I mean, for example consider we have two strongly correlated variables with high VIF and some independent variables. But the model estimates coefficients with acceptable CIs (for example imagine I only care about the sign of the coefficients and the CI does not contain zero), can I trust the CI under these conditions? or the estimated CIs are not valid due to multicollinearity?

Thank you!

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[Reply](#)

Jim Frost says

August 16, 2023 at 11:17 pm

Hi Pourya,

That's correct. Multicollinearity increases the variance but doesn't bias the estimates. So, if the CIs are acceptable (they do not contain zero and are narrow enough to be meaningful for your study area), then you can trust and use them. The common problem is that multicollinearity will widen the CIs enough so that they include zero. If that's not happening, then you're OK. Some researchers might want a more precise coefficient estimates and take steps to address it for that reason. But, yes, the the original CIs are valid even with multicollinearity.

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[Reply](#)

Dileep says

June 28, 2023 at 5:56 am

Thank you for the article, very interesting. I am still mystified by the 3rd point on where multicollinearity can be ignored – but that perhaps is for another day.

I am mainly confused by the example where removing the means from the variables makes the model look better. My thinking is that, the model is exactly as good or as bad as before, except the coefficients are slightly different now (mainly because the rearrangement of linear terms coefficients into the product variable, since mean subtraction affects the product term as well).

Because of this, I don't think the new coefficients for linear terms have the same interpretation, as moving one of the variable by one unit impacts the product term as

well (not just from the multiplication but also due to the presence of linear terms within the product due to mean subtraction).

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[Reply](#)

Jim Frost says

June 28, 2023 at 6:09 pm

Hi Dileep,

Multicollinearity doesn't affect how well the model fits the data. So, if you don't need to understand the role of each IV or their significance, you don't need to fix multicollinearity. It's as simple as that! Of course, there's a lot of studies where the researchers specifically want to understand the role of the variables, so ignoring multicollinearity isn't an option in those cases.

You're not "removing" the means. What you do is subtract each variable's mean from the observed values of the variables. And that only helps fix structural multicollinearity (e.g., polynomials and interaction terms). That process is called centering.

The reason why this works is that it reduces the correlation between the original predictor (x) and any new predictor we create from it (like x -squared or x -cubed). This is because when you subtract the mean from each value, the new mean becomes zero. So, when you square or cube this centered variable, the values are less likely to be directly proportional to the original values, reducing the correlation.

It's like if you were trying to predict a person's running speed, and you had their age and their age-squared as predictors. Younger and older people might run slower (creating a U-shaped curve when you graph it), which is why you'd want to include age-squared. But age and age-squared are closely linked. If you adjust the ages so they are centered around the mean age (so some are positive and some are negative), squaring the centered age doesn't result in the same sort of increasing values as the raw age variable, which reduces the correlation between age and age-squared.

Importantly, centering variables does not change their scale. That's why it doesn't change the interpretations of the coefficients. However, if you standardize the variables, which involves centering the variables and then dividing by the standard deviation, that does change the scale, and hence the interpretation. That's why I recommend centering variables rather than standardizing them even though both approaches will reduce structural multicollinearity.

If you want to confirm this for yourself. Simply take a simple regression model (1 continuous IV) and fit it with and without centering the IV. You'll see that the coefficient remains the same although the constant changes.

In general, I recommend centering continuous IVs anytime you have polynomial and/or interaction terms in your model.

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[Reply](#)

Federico says

May 20, 2023 at 5:54 pm

Thank you! All very clear. I mentioned that more data (larger n of observations) would possibly reduce the negative effect of MC because the coefficients' standard error depend on the inverse of the square root of n so the larger n the smaller the SE and the narrower the confidence intervals get... THANK YOU

Loading...

[Reply](#)

Jim Frost says

May 20, 2023 at 6:00 pm

You're very welcome, Federico!

True, sample size tends to reduce the standard error. However, it might be insufficient when your multicollinearity is bad enough. If time and money are no object and you can collect a larger sample, it could be a way to go. I'd usually recommend using Ridge or LASSO regression, which can handle multicollinearity without needing to gather more data.

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[Reply](#)

Federico says

May 19, 2023 at 7:28 pm

Hello Mr. Frost, congrats on the blog post on multicollinearity. I have some (too many) follow-up questions. Sorry I got carried away but I have been thinking about these matters for quite some time:

a) Given two correlated predictors X_1 and X_2 and the model $Y = b_1 * X_1 + b_2 * X_2 + b_0$, we end up with regression coefficients b_1 and b_2 that are unstable (i.e. if we add/remove a point from the dataset, the coefficients change drastically even if they shouldn't), are statistically not significant (even if the overall model is statistically significant) and have very large standard errors. Large standard errors lead to large confidence intervals CI for each estimated coefficient. But if the estimated regression coefficients b_1 and b_2 are unreliable (the model cannot really untangle and figure out the true unique contribution of each predictor in explaining the variance of Y) why does it matter if the standard errors, hence the confidence intervals, are large or not? The estimated coefficient are really not useful from the start. Also, how large do the standard errors need to be to be deemed "too large"?

b) One often mentioned solution for multicollinearity is to acquire more data (hard to do). But really deflates the standard errors associated to the estimated regression coefficients. However, from an inference point of view, those estimated coefficients remain not accurate, correct? So deflating their standard errors does not seem much progress. As far as the prediction performance of the model, multicollinearity is not an issue at all. It represents an issue only in terms of inference, correct?

c) In a multiple linear regression model, each coefficient is computed via the partial derivative of Y w.r.t. X_1 and represents the change in Y for a unit-change in X_1 while X_2 is kept constant. I guess this interpretation of the coefficient is only true for multiple linear regression models (I see it very hard to interpret this way the estimated coefficients if the model is curvilinear). The calculation to find the coefficient b_1 is essentially the partial derivative. The partial derivative implies that the other predictor X_2 is kept at a constant value (any value really). However, if two predictors are correlated, changing one predictor changes the other, in principle. It is impossible to keep one from changing while the other does. So how does the partial derivative calculation manage to keep the predictor X_2 fixed while X_1 increase by 1? Can it really do that? I am confused. For example, let's consider $Y = 3X_1 - 4X_2 + 3$. How would we interpret those coefficients?

d) It is possible for the sign of a coefficient associated to a predictor to change if we add a new predictor to the model. Does that sign change happen because of collinearity between the predictors? Why a change of sign? That indicates a very radical change of the relationship between Y and the predictor...

e) I know that the calculation of the regression coefficient b_1 involves the correlation between Y and X_1 as well as the correlation between X_1 and X_2 . Is the regression coefficient b_1 , also called partial slope, calculated using what they called the "partial correlation of Y and X_1 " (which partials out X_2)? I have also read about the semi-partial correlation coefficient. Does the semi-partial correlation have any role in the computation of the regression coefficients in multiple linear regression?

Thank you!

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[Reply](#)

Jim Frost says

May 20, 2023 at 4:50 pm

Hi Federico,

Whew! That's quite the list! Here are some quick takes on them.

A) Multicollinearity (MC) causes the large standard errors and wide CIs. That's the heart of MC's problem. It is what makes the coefficients less useful by making them imprecise. It also reduces the power of the t-tests for the coefficients. That's precisely what makes the coefficients less useful or potential useless—that lack of precision caused by MC.

B) Gathering more data won't solve multicollinearity's problems in most cases. If the IVs are truly correlated, more data just means more correlated data. One exception I can think of is if the initial correlation is due to chance. (The IVs are not truly correlated in the population but random sampling error caused of the appearance of that chance in your sample.) In that case, more data might erase that chance correlation. But usually not.

C) The mathematical process for least squares regression doesn't use derivatives. For more information about the calculations, read [Least Squares Regression: Definition, Formulas & Example](#). However, the fact that it is hard to attribute an effect to one IV when there is multicollinearity is what inflates the standard errors. So, yes, that's a problem! You'd interpret the coefficients the same way as usual, but you'd know that the estimates are less precise (larger margins of error).

D) Yes, I show an example of that in this blog post. The standard error is large enough relative to the proximity of the estimate to zero that it can swing the estimate from positive to negative or the other way. It's just a function of being a relatively imprecise estimate.

E) Semi-partial correlation is a related concept to the regression idea of coefficients being the effect while holding other variables constant. Semi-partial correlation is the correlation between X and Y while removing the effect of other variables. It's the same idea but it shows the effects in different ways. Coefficient estimates are unstandardized effects, while correlations are the related standardized effects. For more information, read [Effect Sizes in Statistics](#) where I show pairs of unstandardized and standardized ways of reporting effects.

I hope that clarifies those issues for you!

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[Reply](#)

Lutfi says

December 3, 2022 at 10:46 pm

Hi Jim,

I centered and VIFs decreased significantly yet the marginal effects of the IVs but their average marginal effects (not their regression coefficients but dydx coefficients) remained the same. How did lower VIFs help in this case?

In your sample regression, wouldn't the regression coefficients change anyway after centering unattributable to multicollinearity?

Thanks,

Lutfi

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[Reply](#)

Jim Frost says

December 3, 2022 at 11:58 pm

Hi Lutfi

I'll address the second part of your question and then circle back to the first because it's easier to understand that way.

So, centering a continuous variable simply shifts its values by the mean of that variable. If you picture a scatterplot of a simple model with one X variable and the y variable. If you center the X variable, you simply take the dots on the scatterplot and move them all left or right by a set value. Crucially, the spacing and the overall pattern between the dots remains the same. Consequently, no, you would not expect the coefficient to change because centering doesn't affect the slope. Of course, it affects the y-intercept.

Now onto marginal effects and their average. A marginal effect is the instantaneous effect at a particular data point. Because the pattern of data points remains the same with centering, you'd logically expect the instantaneous marginal effect to be the same at each data point. And, if all the marginal effects are the same, their average will also be the same!

All the above applies to centering because you're shift the x values by a constant amount for all data points. However, if you standardize a variable, which is centering and then dividing by the standard deviation, that changes the spacing between data points and does affect the coefficients and marginal effects.

Centering helps by reducing the variance of the coefficient estimates. Ultimately, that means that your coefficient estimates are likely to be closer to the correct value. These estimates are more precise, producing tighter confidence intervals and greater statistical power.

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[Reply](#)

shobana says

[June 23, 2022 at 1:37 pm](#)

Do you have any thoughts on multicollinearity in models with an independent variable and the square of an independent variable? The article below suggests that multicollinearity is not an issue in this case as we are not interested in knowing the separate effects of the independent variable vs. the squared independent variable on y but rather the overall relationship between the independent variable and y. Do you agree?

<https://statisticalhorizons.com/multicollinearity/>

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[Reply](#)

Jim Frost says

[June 23, 2022 at 5:37 pm](#)

Hi Shobana,

That article says you don't need to worry about multicollinearity when it's caused by the inclusion of "powers or products," or the structural multicollinearity that I refer to in my article.

I disagree with the other article a bit, particularly for the "products" portion. In my article, I show an example with an interaction effect, which is the product of two IVs. The other author is correct that for the product term, the p-value doesn't change. However, for the lower-order terms that comprise the product, the coefficients and p-values do change! In my example, not only does a p-value become insignificant, but the coefficient changes signs, from positive to negative. It definitely affects how you interpret the main effects!

You asked specifically about squared terms, or the "orders" portion. I do think it's less of a concern there. With squared terms, you're more interested in the fact that there is significant curvature for this IV. It's still possible that the p-value and coefficient for the main effect can change values, but if you know the squared term is significant, that's usually enough.

However, fixing this type of structural multicollinearity is so easy and there are no drawbacks to it. Just center the variables. There's really no reason not to do that. Consequently, I recommend centering for models with powers and products. Additionally, by removing the structural multicollinearity, you can more easily identify the more problematic data multicollinearity.

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[Reply](#)

shradha says

[April 23, 2022 at 10:37 am](#)

Hi Jim, really appreciate the way you explain the issues of multicollinearity. I have a question. For a panel regression analysis, if one variable (say X) in a multivariate regression has a high VIF around 15, rest all independent variables have moderate VIFs, is it required to remove the variable X from the equation? Thank you.

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[Reply](#)

Jim G says

April 8, 2022 at 5:26 pm

Hi Jim. Thanks for the interesting article. I am trying to understand this concept better by replicating your results but I have been unable to duplicate your VIFs using the centered data. My understanding is that the VIF for an independent variable, Y_1 , is equal to $1/(1-R^2)$ where the R^2 is generated by the equation whereby Y_1 is regressed against all of the other independent variables. For the uncentered data, I got the identical answers that you did for the VIFs. So far, so good. When I regress the centered data to calculate the VIFs for the centered variables (e.g., regress % Fat S against Weight S, Activity S and % Fat* Weight), however, I got the identical VIFs as those for the uncentered data. So, in the % Fat S vs. the other three independent (centered) variables example, I got an R^2 for the regression of .933 which corresponds to a VIF of 14.93 (identical to the VIF for the uncentered % Fat. Am I missing something here or doing something incorrectly? Thanks in advance.

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[Reply](#)

Jim Frost says

April 8, 2022 at 7:55 pm

Hi Jim,

Your understanding is correct. I'm not sure why the calculations are not working out correctly. Centering will reduce multicollinearity for higher-order terms as shown. I can only suggest that you carefully check through your procedures and calculations. Be sure that you are centering all continuous IVs and that you only use the centered variables in the analysis. I wish I could tell you more precisely how to get the correct answer, but I can guarantee that centering will lower those VIFs in the example.

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[Reply](#)

Mary says

March 5, 2022 at 6:09 am

Hi Jim

Thanks for the wonderful information

However I have a question, am performing a Heckman two stage model but it shows that dependent variable is collinear with the constant

What should I do???

Thank you.

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[Reply](#)

Jim Frost says

March 6, 2022 at 3:27 pm

Hi Mary,

I'm not particularly familiar with the Heckman two step method. However, I'm not sure how a variable can be correlated (collinear) with a constant? Correlation is when two variables tend to vary in unison. Your DV can vary but the constant cannot, so correlation seems impossible unless it relates to the two-step nature of the process somehow. But I don't know enough to be able to say.

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[Reply](#)

Dr. MM Ali says

February 2, 2022 at 4:44 am

When does the scatter sometimes has vertical clustering?

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[Reply](#)

kanchan singh says

January 27, 2022 at 9:00 am

Thanks a lot for your valuable post. I am greatly benefitted by this. It is an asset for me to teach my students. In social science research there are several variables which remain inclusive and influence the dependent variable to a large extent. Multicollinearity remains a puzzle to be resolved. With your inputs, I am in a position to explain the the cause and effect.

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[Reply](#)

Jack says

December 21, 2021 at 12:59 pm

Jim,

This has helped me very much with my coursework.

Just to clarify, the model I am using regresses Y on x_1 & x_2 where the error term is correlated with x_2 .

I have to investigate certain econometric issues relating to the intercept and x_1 .

Given that the error term is only correlated with x_2 , will my estimates for the intercept and x_1 remain unbiased?

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[Reply](#)

Jim Frost says

December 21, 2021 at 3:38 pm

Hi Jack,

Just to clarify, multicollinearity is correlation between the X variables in your model. It is not correlation between an X variable and the error term. To learn more about what you describe, read my post about [OLS assumptions](#) and pay particular attention to assumption #3. There are differences between these two conditions, which affects how you should proceed.

As I discuss in this article, multicollinearity only affects X variables that are highly correlated. Technically, multicollinearity is not a bias because it inflates the variance but does not systematically shift the coefficients in one direction or the other. Unfortunately, the type of bias your model might contain is not like that. It can affect the other variables. Think of the regression equation. If one coefficient is biased to high, the other coefficients have to adjust to produce the average Y value.

You really should address that correlation between X2 and the error. In return, you'll get a better model!

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[Reply](#)

reyhan says

December 3, 2021 at 1:14 am

thanks alot of information

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Reply

PRW says

November 7, 2021 at 1:30 pm

Thank you for this post. Very helpful. Do you have a reference for the VIF values?

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Reply

Jim Frost says

November 8, 2021 at 1:05 pm

Hi Patrick,

VIFs starting at 1 and increasing upwards without limits is based on how it is calculated. For more information, read my post about [Variance Inflation Factors](#).

As for the cutoff points, there are no universally accepted values. Most statisticians will tell you that problems start when VIFs are greater than 5 or greater than 10. I have two textbooks at hand. The one that I reference partway through this post, *Applied Linear Statistical Models*, suggests that problems start when $VIFs > 10$. However, the other textbook I have, *Using Econometrics: A Practical Guide* by Studenmund, uses $VIFs > 5$. You'll find references both ways. In fact, some statisticians link the cutoff value to the number of IVs in your model. It's easier to get higher VIFs with more IVs.

Personally, I use the cutoff of 5. Based on experience, that seems to be where problems frequently start.

I hope that helps! There really is no definitive answer. It's just that as VIFs grow, so do the effects of multicollinearity. At some point, those effects become problematic. Some of the determination depends on the nature of your data and what you want to do. These days, it's easy enough to perform Ridge or LASSO

regression, which are better equipped to handle multicollinearity if you have any doubts.

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[Reply](#)

Paulo Rodrigues says

October 25, 2021 at 6:49 am

Hi Jim,

Is it possible for you to explain me how to use the PCA to solve the problem of multicollinearity in a regression?

Thank you

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[Reply](#)

Jasbir Singh says

October 25, 2021 at 1:33 am

Hello Jim,

Thank you for this blog – it was really interesting.

1 question: If a variable is not having multi-collinearity should we standardize it too – along with other variables having multi-collinearity?

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[Reply](#)

Jim Frost says

October 25, 2021 at 11:00 pm

Hi Jasbir,

If a variable does not have problematic levels of multicollinearity, then you don't need to standardize (or center) it. I center the variables which gives you the same benefits as standardizing without changing the interpretation of the coefficients. However, if you do center all the continuous variables, you get the added benefit that the constant equals the mean DV when all IVs are at their mean, which can be a useful thing to know. So, centering all continuous IVs is not required but it can be helpful!

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[Reply](#)

Susan Hardy says

October 20, 2021 at 8:32 pm

I am trying to use your data set for this post in a class I teach (of course I am sourcing you). What are the units for activity? Also how was activity level measured to be in the thousands. Steps per day?

Is the units for bone density g/cm^2 ?

Thank you for a great example—and especially for the data set so we can create your output, and I can teach how to program it.

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[Reply](#)

Jim Frost says

October 22, 2021 at 1:15 am

Hi Susan, I collected this data about 20 years ago. So, I'll need to look through my records to refresh my memory!

I'm so glad it's been helpful for you!

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[Reply](#)

RABIA NOUSHEEN says
June 28, 2021 at 3:36 am

Hi Jim

I want to ask if main effects (Individual predictors) show multicollinearity greater than the interaction terms in regression analysis then is it sensible to remove the individual terms from the model?

Thank you for your help

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[Reply](#)

Taiwo says
June 15, 2021 at 8:15 am

What an awesome simply work. Keep it up

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[Reply](#)

Jim Frost says

June 19, 2021 at 4:18 pm

Thanks, Taiwo!

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[Reply](#)

Joblin Omari says

May 12, 2021 at 1:00 am

This is really helpful, for the first time i have seen statistics simply explained

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[Reply](#)

Bayarbaatar says

May 6, 2021 at 4:06 am

Dear Jim,

Thank you for your great explanation about multicollinearity. It was really understandable and made me more confident.

So I would like to ask you about what software (R or SPSS) is best and easy to use to detect VIF on windows operation system

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[Reply](#)

Amanda says

April 30, 2021 at 10:45 am

"The interpretation of a regression coefficient is that it represents the mean change in the dependent variable for each 1 unit change in an independent variable when you hold all of the other independent variables constant."

That really helped me understand why multicollinearity can be a problem. Thank you for this well-written post!

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[Reply](#)

Jim Frost says

April 30, 2021 at 3:03 pm

Hi Amanda, you're very welcome! So glad to hear that it's been helpful.

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[Reply](#)

muhammad ali raza abid says

April 24, 2021 at 2:35 am

Dear Jim I am pretty thankful that you are sharing your experience. I did a big mistake in centering the variables! Felt shameful for that! What I did is that I took the mean of the dataframe (including all IV's) and then subtracted the original ones from the mean which increased VIF.

However Good news for me! I want to tell you that actually VIF's are lowered after subtracting the mean from the variable 1st and then creating the polynomial terms from that variable. However the lowered VIF's are still ranging from 100-600. Before

they were in the order of 1000's. This is well in agreement with the fact you mentioned before that the question is not that VIF's are lowered or not after centering but the question is how much VIF's are reduced. So in my case they are not reduced enough probably! I have also noted your comment that 5 degree polynomial might be doing overfitting. Actually my purpose for creating 5 degree polynomials is nothing but to improve the prediction accuracy which I am getting from the model but unfortunately at the expense of high VIF's which are though now reduced a bit after centering but still multicollinearity is there. Any suggestions regarding this case study will be highly appreciated please. Thanks!

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[Reply](#)

[muhammad ali raza abid](#) says
April 22, 2021 at 2:52 am

Dear Jim, I have rechecked again and found that VIF increases by centering the variables i.e. taking the mean and subtracting them , again testing for VIF. I am also surprised to see my results! Any suggestions to further investigate this issue will be highly appreciated! Actually in my case I have 19 IV's. I found the most important IV and created 5 degree polynomials for that variable. This process significantly increased by Adjusted R Square (about 10%). But the issue of multicollinearity remain there! Thanks for considering my question.

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[Reply](#)

Jim Frost says
April 23, 2021 at 11:27 pm

Hi Muhammad,

I'm still at a loss. However, if your fitting a 5th degree polynomial is extremely unusual. That's almost definitely overfitting your data unless there is a very

particular reason that your data requires it. It might also explain why centering isn't helping the VIFs! In practice, I almost never see even a 3rd degree polynomial. Usually a quadratic term is the highest in practice.

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[Reply](#)

Batru Wolde says

[April 21, 2021 at 11:58 am](#)

Hi Jim Thank you for your contributions. Really the material is help full I used it for my research work. Keep it up

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[Reply](#)

[muhammad ali raza abid](#) says

[April 21, 2021 at 4:46 am](#)

Hello Jim

Thanks for this wonderful article. In my case I intentionally created polynomial features and observed that my linear regression model performance is significantly increased. However at the same time as you mentioned I observed that the polynomial features have high correlation with the feature from which they are created and also high VIF values of the order of 1000. Then I tried to do the same as mentioned here i.e. calculated the mean and then subtracted mean from the features to create a new data frame. Then I used that data frame to calculate VIF and correlations. But I observed that VIF values actually increases a bit instead of decreasing, the correlation values on the other hand remains the same as before. Could you please suggest something here! Does v.high values of VIF i.e. the order of 1000 means something abnormal etc?

Although I did check the p-values and they are well below 0.05.
Looking forward to your kind advice on this. Thanks

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[Reply](#)

Jim Frost says

April 21, 2021 at 2:00 pm

Hi Muhammad,

I have to be honest, I've never heard of a case where centering the variables actually increased the VIFs! Are you sure the centering process was implemented correctly? It should reduce the VIFs and the main question is whether it reduces them by a sufficient amount. Again, I've never seen a case where it increase them and that really shouldn't happen. After you center all of the variables, the mean of each variable should be virtually zero.

I will need to think about this one to see if there are cases where VIFs could increase.

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[Reply](#)

David says

March 14, 2021 at 4:47 pm

Dear Jim

this post is early amazing and easy to understand. but I have collinearity problem with my data. my data is a univariate time series of the load consumption(with hourly resolution), I use the sliding technique to make the input vector of NN that includes the lag variables. I want to solve this with two-step correlation analysis, the first one that is

the easiest part and calculate the relevancy (correlation with the target, in which candidate variables with higher correlation value than threshold are retain) but the second one is problematic, and the idea(according to one of journal's paper) is cross correlation in which ' if the correlation index between any two candidate variables is smaller than a pre-specified value cor_2 , then both variables are retained; else, only the variable with the largest correlation with respect to the output is retained, while the other is not considered any further'. but according to the correlation matrix it is not possible because all nearly located lags are highly correlated it means that according the assumption of the paper, variables should retain because their correlation index is lower than the second threshold and should be discard because the high correlation according to the recent lags.

for example: $t-1, t-2, \dots, t-24, \dots, t-48 \rightarrow t-2$ and $t-48$ should be retain because they are not correlated but $t-2$ should be discard because it is highly correlated with $t-1$.

what is your opinion ? isn't it a contradict? how can I solve it in general and how can I solve it by correlation analysis?

Bests

David

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[Reply](#)

YJ says

February 13, 2021 at 11:16 pm

Hmm, okay, I think I can understand that, it targets structural multicollinearity rather than data multicollinearity.

I'm trying to think through the implications of this technique then of a regression model that exhibits multicollinearity, but not structurally. My intuition is that the difference between $y = b_0 + b_1x_1 + b_2x_2$ and $y = b_0 + b_1c(x_1) + b_2c(x_2)$ is that the coefficients remain the same, but b_0 increases by $\text{mean}(x_1) + \text{mean}(x_2)$. But honestly, neither here nor there, since that's a data multicollinearity problem! I should just throw PCA at it.

Thanks for responding, by the way, I do appreciate the effort and work you put into these posts; it's hard to describe statistics as straightforwardly as you do!

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[Reply](#)

Jim Frost says

February 14, 2021 at 12:06 am

Hi YJ,

Yes, centering the variables changes the constant. But, it does reduce multicollinearity, which helps you accurately determine the statistical significance of your predictors. Multicollinearity reduces the ability to detect significant effects.

And, thanks! I do strive to explain statistics in a straightforward yet completely accurate manner, which can be difficult to both sometimes!

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[Reply](#)

YJ says

February 9, 2021 at 8:13 pm

Jim,

I'm having a really hard time intuiting how standardising the vars would reduce structural multicollinearity without it being a hack in the maths.

For instance, if we have two vars, x_1 and x_2 , and we know them to be collinear, we would expect to see that relationship in the plot of x_1 against x_2 . In effect we'd expect to see a high R^2 between the two, instead of the 0 that denotes orthogonality. But $\text{mean}(x_1)$ and $\text{mean}(x_2)$ are both constants, so subtracting their means from the observed values would only translate the datapoints left and down: I don't see how that would reduce the R^2 , or the corr coefficient, $s(x_1)$ and $s(x_2)$ are still correlated to the same extent.

In thinking about it that only thing I can think in how it addresses that collinearity issue is that it percolates through to the actual regression, and “reduces” the effect this collinearity has on the dependent var by altering x_1 and x_2 ’s coefficients while decreasing x_1 and x_2 ’s numerical values, by essentially adding two more constants to the regression, so instead of $y = b_0 + b_1x_1 + b_2x_2 + e$, regressing instead on $y = b_0 + b_1x_1 + b_2x_2 - b_1.\text{mean}(x_1) - b_2.\text{mean}(x_2) + e$. So we kind of leave the collinearity there, but add more “secondary” constants to account for it. Is this correct?

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[Reply](#)

Jim Frost says

February 11, 2021 at 4:49 pm

Hi YJ,

It reduces the extremeness of the products. Say you’re talking about an interaction $X_1 * X_2$. By centering the values, they’re all close to zero. That reduces the impact of the multiplication on larger values. It’s an accepted and valid solution for reducing structural multicollinearity.

Note that you don’t need to standardize (center and divide by the standard deviation). You only need the centering portion. However it works using either method. The advantage of centering only is that it doesn’t change the interpretation of the coefficients.

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[Reply](#)

RABIA NOUSHEEN says

February 6, 2021 at 3:04 am

Hi Jim

It is a very useful article. Thanks for that.

I am recently facing multicollinearity in my data. Can you please share R codes to standardize the categorical variables and their regression. My variables are categorical like Toxicant type(4 levels), Exposure time(4 levels) and concentration(4 levels).

Loading...

[Reply](#)

Jim Frost says

February 6, 2021 at 11:46 pm

Hi Rabia, unfortunately R isn't my strong suit.

Loading...

[Reply](#)

Louise says

December 10, 2020 at 9:40 am

Thank you for a really clear and useful explanation. I have what may be a stupid question (apologies if it is!). I'm doing some analysis regarding a predictive test which generates complex spectroscopic data. Previous analysts have used binomial LR to combine related variables (spectroscopic readings of the same tissue at different frequencies, ie same subject and measurement obtained at more or less same time) to produce a summary measure which is then used to predict a dichotomous clinical outcome. The combined variables are obviously not independent and consequently have severe multicollinearity. But the output probabilities are only being used for prediction. I'm therefore unsure if this is a completely invalid approach or not! Much of the literature in this field is engineering/computing based and quickly becomes incomprehensible (as someone not in those fields!).

Thanks again for a useful article!

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[Reply](#)

Jim Frost says

December 10, 2020 at 10:15 pm

Hi Louise,

This approach sounds like a good one to me. In my post towards the end, I have three bullet points for potential solutions. The second bullet states, Linearly combine the independent variables, such as adding them together. That's essentially what the researchers are doing. Combining highly correlated variables together. This removes the multicollinearity because you can have correlation between predictors when there is only one predictor! Well, you can have other predictors in the model, but this process combines the correlated predictors into a combined measure.

The downside of this approach is that you don't know the role of each predictor. However, if the primary goal is to predict the outcome rather than understanding the role of each predictor, that's not really a problem here.

So, I don't see anything inherently wrong with the approach for this case.

Thanks for writing with the interesting scenario!

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[Reply](#)

Fernanda Hornos says

December 1, 2020 at 10:57 am

Hi Jim! We have two IV that are highly correlated (-0.86) and our VIF is 3.90. Do you think that we continue to trust the data or do we have to make a PCA?

Loading...

[Reply](#)

Jim Frost says

December 1, 2020 at 11:34 pm

Hi Fernanda,

I don't think that's a problem. It does sound like a high correlation but the VIF is below 5. You should be good. With just two IVs, you'd need an even higher correlation to be a problem. I did some calculations and found that the VIFs won't reach 5 until the correlation reaches an absolute correlation of 0.894 when you're dealing with only two IVs. So, you're getting close but you should be OK. You can stick with regression! Thanks for writing!

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[Reply](#)

Danny Data says

November 29, 2020 at 9:53 pm

Jim,

I was working on finding the VIF in your dataset using Python to follow along with your post. I ran the following and my initial VIF for the Independent Variables was much higher:

```
import pandas as pd
from statsmodels.stats.outliers_influence import variance_inflation_factor

df = read_csv("MulticollinearityExample.csv")

df['%Fat*Weight kg'] = df['%Fat']*df['Weight kg']

x = df[['Activity', '%Fat', 'Weight kg', '%Fat*Weight kg']]
```

```
vif_data = pd.DataFrame()
```

```
vif_data['feature'] = x.columns
```

```
vif_data['VIF'] = [variance_inflation_factor(x.values, i) for i in range(x.shape[1])]
```

```
print(vif_data.round(2))
```

feature VIF

0 Activity 7.48

1 %Fat 53.21

2 Weight kg 66.93

3 %Fat*Weight kg 26.73

After applying initial standardization as noted, my VIF matched your initial VIF from your example:

```
for iv in ['Activity', '%Fat', 'Weight kg', '%Fat*Weight kg']:
```

```
df[iv] = [x - df[iv].mean() for x in df[iv]]
```

feature VIF

0 Activity 1.05

1 %Fat 14.93

2 Weight kg 33.95

3 %Fat*Weight kg 73.06

Applying it one more time I got your same end results:

feature VIF

0 Activity 1.05

1 %Fat 3.32

2 Weight kg 4.75

3 %Fat*Weight kg 1.99

Any idea why this occurred? Thanks for your help!

Loading...

[Reply](#)

Jim Frost says

December 2, 2020 at 12:11 am

Hi Danny,

I'm really puzzled as to what is happening. Unfortunately, I don't have sufficient Python knowledge to help there. However, I have double-checked the my VIF values and they're correct. You can try calculating the VIF yourself to see where the problem lies. Take one of the IVs and use that as the DV. Then take the remaining IVs and regress that on the new DV (the former IV). Don't include the original DV at all. From that regression take the R-squared and calculate the VIF: $1 / (1 - R^2)$. Perhaps that will point you in the right direction.

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[Reply](#)

Heather says

October 30, 2020 at 1:15 pm

Hi Jim,

Thanks so much for your response – this is really helpful and makes much more sense now! I checked the residuals following your instructions on the relevant blog page and all look fine – thank you for recommending this.

Best wishes,
Heather

Loading...

[Reply](#)

Jim Frost says

November 1, 2020 at 10:12 pm

Hi Heather,

You're very welcome! It's definitely good news that the residuals look good. That means there's no obvious problems!

Loading...

[Reply](#)

Prince says

October 15, 2020 at 5:31 pm

When doing multiple regression when does multicollinearity matter (i.e. worth examining) and when does it not?

Loading...

[Reply](#)

Jim Frost says

October 16, 2020 at 3:46 pm

Hi,

You're looking at the right article to answer your question. You'll find your answer in the section titled, "Do I Have to Fix Multicollinearity?" It doesn't make sense for me to retype it here in the comment but all the details are there! If anything is still not clear, please don't hesitate to ask!

Loading...

Reply

Heather says

October 9, 2020 at 7:49 am

Hi Jim,

Thanks so much for the post, it's very helpful. Do you have any advice on how to interpret a model which is significant overall, but non of the predictors reach significance? The highest VIF i have is 3, for two of the seven predictors which, after reading your blog, doesn't seem like something to worry about. I'm generally interested more in the predictors (i.e., which might predict the outcome most strongly). A significant overall model is great, however i'm not sure how to interpret the fact that non of the predictors are significant, or whether this indicates another problem in the model since most advice I am finding online is that this would normally indicate a problem with multicollinearity – but it seems this isn't the problem because VIFs aren't very big?

Thank you!

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Reply

Jim Frost says

October 13, 2020 at 2:42 pm

Hi Heather,

This situation usually happens when your model is borderline significant. I describe your situation in more detail in my post about [the overall f-test of significance](#).

I agree that multicollinearity is not likely a problem given your VIFs. But, you're correct, when multicollinearity is present, that can happen.

What it probably means is that there is just enough evidence to conclude that your model as a whole predicts the DV better than just using the mean. However, there's insufficient evidence to link it to any individual IV. Technically, the significant overall f-test indicates that all seven of your coefficients are unlikely to equal zero simultaneously. However, there is not enough evidence to identify a specific coefficient that is different from zero. Being different from zero is equivalent to being statistically significant for these tests.

I explain this in more detail in the post that I link to above.

Basically, your model is borderline significant and there's just not enough information to pin down the significance more specifically to specific variables. That's how I'd read it. Another alternative is that your model isn't accounting for curvature or interaction terms. Check those residual plots to look for obvious patterns that you can correct.

I hope this helps!

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[Reply](#)

[John Grenci](#) says

October 4, 2020 at 6:24 pm

Hi Jim, as always much appreciate the articles, and this one in particular is quite interesting. Count me among the many who take issue with the third item listed above. I know that some things in mathematics are not perhaps as intuitive as others... but what if you have two IV's that are correlated. Assume that both have a low p value and for illustration, let's assume the coefficient of variable A is 3 and variable B is 5. Thus, we predict the DV by $3a+5b$. However, is it not true that if we cannot trust the p values, then we cannot trust the coefficients, and thus cannot trust the prediction? thanks John

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[Reply](#)

Hans says

October 1, 2020 at 7:29 am

Hi Jim,

I'm running a GLM model. I have used VIF and remove the covariate that was >10 .

I checked for correlation and there is still some covariates that is correlated. I remove one of the correlated covariate and check the BIC value, if it decrease I continue with it. At the same time I'm aware on statistical significancy. But two of my covariates is correlated(0.62), but when I remove one of them from the GLM model BIC increase. All of my VIF values is below 2.5. What can I do?

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[Reply](#)

Jim Frost says

October 1, 2020 at 4:44 pm

Hi Hans,

In some cases, handling multicollinearity can be very difficult to remove. There are times when the treatment is worse than the problem.

A correlation of 0.62 isn't necessarily problematic. Give the VIFs priority over the pairwise correlations. If the VIFs are fine (less than 5), I wouldn't worry about the pairwise correlations.

If removing a correlated predictor improves the goodness-of-fit, that's also a net gain. A simpler model and an improved fit. That sounds promising!

If you really need to include a correlated predictor that you otherwise would need to exclude for the reasons above (may be you need to estimate its effect and test its significance specifically), I'd try Ridge or LASSO regression. These forms of regression analysis are better able to handle multicollinearity. I write a little bit about both of them in my post about [Choosing the Correct Type of Regression Analysis](#). Look for Ridge and LASSO regression in that article to see what it can do

for you. I don't have an article that shows them in action, at least not yet. But, it would be an avenue for you to explore. Using these forms of regression, you'd leave the correlated predictors in the model.

I hope that helps!

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[Reply](#)

purva garg says

[September 14, 2020 at 3:17 am](#)

Hi! thanks for such informative article. In sources of multicollinearity does high leverage values also come? I can't understand the intuition behind it. it will be helpful if you could explain how high leverage values are a source of multicollinearity?

Loading...

[Reply](#)

Pavithra says

[September 13, 2020 at 12:27 pm](#)

Hi sir, I have one doubt in my research totally i taken 16 variables in that 11 variables are collinearity . i did structural model equation with 11 variables. is it is good or not sir. kindly give valuable suggestion.

Loading...

[Reply](#)

Jim Frost says

September 13, 2020 at 10:13 pm

Hi Pavithra,

You need to check the VIFs as I show in this post. See what those values are to determine whether you have problematic levels of multicollinearity. If you do, you might need to use something like Ridge or LASSO regression.

Loading...

[Reply](#)

Angela Kitali says

August 21, 2020 at 11:46 am

Hello Elif. You definitely can. The approach is just a little bit different from the usual way we compute VIF. Instead of using VIF, a generalized variance inflation factor (GVIF), proposed by Fox and Monette (1992), is used to check for correlation when all variables are categorical. Refer to this document for more information {Fox, J., & Monette, G. (1992). "Generalized collinearity diagnostics". Journal of the American Statistical Association, 87(417), 178-183.}.

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[Reply](#)

Sauddi Syamier says

June 17, 2020 at 4:52 am

Hi Jim,

Thank you for a very insightful post!

I am regressing between the relationship of net profit(dependent variable) with revenue, cost of sales and expenditure (independent variables). My analysis shows R-square =1, all the coefficients are 1 and p-values are all 0 which is very weird.

I found out that multicollinearity exists between all of the independent variables as regressing them with each other gives me a high R-square figure (>0.95) and significant p-values so I think this is a case of multicollinearity in the data set.

I noticed that there is not a lot of regression out there on accounting data/equation so im not sure if this type of regression is possible or not. It would be best if you could share with me your professional opinion regarding this matter.

Thanks,
Sauddi

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[Reply](#)

Elif says

June 16, 2020 at 10:24 am

Hello, firstly, thank you very much for this very helpful post. I have categorical independent variables, can i use vif to determine multicollinearity?

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[Reply](#)

Jim Frost says

June 16, 2020 at 3:35 pm

Hi Elif,

Unless your software uses a specialized VIF that I'm unaware of, VIFs are only for continuous variables. You might need to use something like Cramer's V to assess the strength of the relationship between categorical variables.

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[Reply](#)

David de Visser says

[June 6, 2020 at 4:09 am](#)

Hi Jim. Thanks a lot in helping me out here! Your articles have been very helpful and I especially appreciate the fast responses. Thanks!

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[Reply](#)

David de Visser says

[June 5, 2020 at 3:14 am](#)

Hi Jim,

Thanks a lot for the fast response. The pairwise correlation is 0.3629 between X1 and X4, which is significant in my data-set with 136 observations.

The VIFs are as follows:

X1: 1.27

X2: 1.08

X3: 1.82

X4: 2.03

All of them are well below five, so I would think that there is no problem of multicollinearity. Or is there still another way through which the significant correlation between X1 and X4 could bias my results?

The explanation on statistically significant and practically meaningful is also interesting and helpful. Can I see the VIF as 'a sort of' test of whether a significant correlation is practically meaningful and causes problems?

Thanks again! David

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[Reply](#)

Jim Frost says

[June 5, 2020 at 7:57 pm](#)

Hi David,

Those values are not problematic at all. The VIFs are all good. The correlation is fairly weak. It's statistically significant, so your sample data support the hypothesis that the variables are correlated in the population. However, that doesn't mean it's a strong correlation. So, no need to worry about multicollinearity!

VIFs are actually a more sophisticated way of checking for correlation than pairwise correlations. Imagine that your X1 has a small correlation with X2, X3, and X4. Each of those pairwise correlations are moderate and no big deal on their own.

However, now imagine all those correlations result in those three variables (X1, X2, X3) collectively explaining most of the variability in X1. Basically, you can use X1 – X3 to predict X4. VIFs can check for that scenario. VIFs regress each X on the remaining set of Xs. $VIFs = 1/(1-R^2)$. So, as R-squared of those regressions increase, so does the VIF.

Now, R-squared measures the strength of that relationship. So, ultimately, VIFs measure the strength of the relationships between each predictor and all the remaining predictors. If you plug in R-squared values into the VIF formula, you'll find that an R-squared of 80% produces a VIF of 5, which is where problems start. Your variable X4 has a VIF of 2.03, which means that if you regress X1, X2, and X3 on X4, that model will have an R-squared of about 50%. You can try that with your variables if you want. If you have more predictors than those listed, you'd need to include them as well.

If you have only two predictors X1 and X2 in your model and they have a Pearson's correlation of 0.9, that equates to an R-squared of 81%, which is again right around where you get VIFs of 5. But, VIFs are better than correlations because they incorporate all the relationships between the predictors, not just one pair at a time.

So, yes, that's a long way of saying that VIFs measure the practically meaningful strength of the relationships between all the predictors.

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[Reply](#)

David de Visser says

[June 4, 2020 at 3:50 am](#)

Hi, first of all thank you for your explanation on multicollinearity. At the moment I am struggling with a regression analysis. I am trying to find the effect of X1 on Y and based on literature I should include control variables X2, X3 and X4. However, when looking at the pairwise correlations between all my independent variables I discovered a statistically significant correlation between X1 and X4. However, when looking at the VIF of the regression model I do not see any problem. So, I would say there is no problem of multicollinearity. I am wondering whether the significant pair wise correlation between X1 and X4 still causes a problem in my regression and whether I should exclude X4 based on the correlation?

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[Reply](#)

Jim Frost says

[June 5, 2020 at 12:33 am](#)

Hi David,

How high is the correlation between X1 and X4? And what are the VIFs for all your variables?

One thing to keep in mind is the difference between [statistically significant and practically meaningful](#). In this context, practically meaningful is a correlation high enough to cause problematic levels of multicollinearity. However, just because a

test result (correlation in this case) is statistically significant, it doesn't necessarily mean it is practically significant.

If your VIFs are low enough, then my guess is that the correlation is statistically significant (you can be reasonably sure it exists in the population) but it's not large enough to cause problems with multicollinearity. The VIFs for each IV are the key statistics to assess here.

In a nutshell, it sounds like multicollinearity is NOT a problem when you include all your x variables.

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[Reply](#)

Julien Mainguy says

[May 29, 2020 at 11:59 am](#)

Hi Jim, thanks for these precisions. I will certainly look into the LASSO and Ridge regression.

In response to your question, I did use previously the approach I've proposed for a published MS of my PhD thesis (see Mainguy et al. 2008 Age- and state-dependent reproductive effort in male mountain goats, *Oreamnos americanus*. Behavioral Ecology and Sociobiology, 62:935-943) in which the proportion of time spent in reproductive behaviours by males was investigated according to age and age-specific body mass within a same model – with age-specific body mass still explaining a significant part of the variation observed in the dependent variable.

And thanks again for your blog, I've found very interesting and useful information in it.

Regards.

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[Reply](#)

Jim Frost says

May 29, 2020 at 4:11 pm

Hi Julien,

Thanks for the extra information. I will have to look into that approach. Honestly, I don't fully understand it. Unless the residuals from the other model are not random, which is usually a bad thing, I'm not quite sure how random residuals would explain variation in a DV. However, I'm glad it helped you out with your thesis!

I can say that using LASSO or Ridge are more common approaches.

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[Reply](#)

Julien Mainguy says

May 27, 2020 at 12:43 pm

Just a quick comment first: congratulations Jim for this very clear and well-written blog about multicollinearity. Well done. Seriously.

And, second, if I may suggest an additional potential solution to deal with multicollinearity: in a model where $Y \sim X_1 + X_2$, but X_1 and X_2 are quite correlated such that one or both VIF values are > 5 , then one could decide to look at $X_2 \sim X_1$, found the predictive equation between these variables and used the residuals of X_2 on X_1 in the model " $Y \sim X_1 + (X_2 \text{ corrected for } X_1)$ ". As an example, if the probability to lay a clutch (or not) for a female bird is examined according to her AGE and BODYMASS, two "independent" variables that are in fact correlated, then one could rather use as independent variables AGE (again) and AGE-CORRECTED BODYMASS in the analysis, as the residuals of BODYMASS against AGE should no longer be correlated in a problematic manner. This should allow to test both variables in a same model without multicollinearity issues. Would you agree with this potential solution?

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[Reply](#)

Jim Frost says

May 27, 2020 at 5:05 pm

Hi Julien,

I'm not familiar with that technique. I'd have to think about it. Offhand, I'm not sure that I'd refer to it as age corrected body mass using your example. I agree the residuals would not be correlated but they're just the random error left over from that model. The random error from one model shouldn't explain anything in another model. If it does, the residuals aren't random. But, I'd have to think it through. Have you tried this method? If so, how did it work out?

I'd recommend using LASSO or Ridge regression as I describe in my post about [choosing the correct type of regression analysis](#). I hope to write a blog post particular about those techniques at some point!

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[Reply](#)

Collin M says

May 22, 2020 at 7:08 am

thanks Jim for ur help.

Now should I take it that multicollinearity can not exist significantly if interaction is absent?

And if so, can I have a decision about the existence of multicollinearity based on whether the interactions are significant in the ANOVA rather than using VIFs.

In other words, can I conclude on existence of multicollinearity based on the P-value of the interaction terms in the ANOVA?

thank you

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[Reply](#)

Jim Frost says

May 22, 2020 at 1:34 pm

Hi Collin,

Multicollinearity can definitely exist when there are no interactions. If the IVs are correlated, then multicollinearity exists. No interaction effect is required. However, interaction terms create multicollinearity unless you center the variables.

Additionally, if you include an interaction term and the p-value is not significant, the interaction term still creates multicollinearity. It's the fact that the interaction term uses the product of two or more variables that causes that type of multicollinearity rather than the significance of the term.

Just keep in mind that interaction terms are just one possible source of multicollinearity, not the only source.

Don't use p-values to assess multicollinearity. P-values don't assess multicollinearity so it's not the right tool to use. Use VIFs as I show in this post.

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[Reply](#)

Collin M says

May 21, 2020 at 12:34 pm

hello Jim, thanks 4 ur great efforts in simplifying statistics and regression analysis. But am confused about the difference between multicollinearity and interaction of the independent variables, do they mean the same thing? And can they occur at the same time in multiple independent variables. I will be glad if u help me out on that. thank you

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[Reply](#)

Jim Frost says

May 21, 2020 at 2:35 pm

Hi Collin,

That's a great question. I've heard similar confusions before about this exact question.

Multicollinearity and interactions are different things.

Multicollinearity involves correlations between independent variables.

Interactions involve relationships between IVs and a DV. Specifically, an interaction effect exists when the relationship between IV1 and the DV changes based on the value of IV2.

So, each concept refers to a different set of relationships. Within the IVs (multicollinearity) or relationships between IVs and the DV that change based on another IV.

And, yes, they can occur at the same time. In fact, when you include an interaction term, it usually creates multicollinearity. The interaction term itself is the product of at least two IVs. Consequently, that term is correlated with those IVs and thereby creates multicollinearity. However, as I write in this post, centering the variables helps reduce that type of multicollinearity.

I hope that helps!

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[Reply](#)

Prabina says

May 21, 2020 at 8:53 am

Hi Jim,

Is there any condition at which regression coefficient is statistically significant in univariate logistic model, but not correlated with each other. By this what I mean is can I use one of the categorical IV as a dependent variable to check for collinearity. I would assume that if the regression coefficient is significant, then they are correlated.

Thank you,

Prabina

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[Reply](#)

Jim Frost says

May 21, 2020 at 10:48 am

Hi Prabina,

When a regression coefficient is statistically significant, and assuming there aren't other problems with the model that is biasing the coefficient, that indicates there is a relationship between the IV and DV. So, there is some type of correlation between the two. In your case, it's not the more familiar Pearson's correlation because you're not dealing with two continuous variables. Instead, you've got a categorical and a binary variable. Still, there is a correlation between them. Your model is telling you that if you know the category for the IV, it gives you information about value for the binary variable.

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[Reply](#)

MAHESWAR SETHI says

April 3, 2020 at 11:40 am

In a model with 4 independent variables, when there is multicollinearity problem among two independent variables. Is it necessary that such two variables need to be centered to remove multicollinearity problem or centering any one variable can remove the multicollinearity problem?

Further, centering any variables requires all the independent variables to be centered even though they don't suffer from multicollinearity problem? As it shown in the example that due to multicollinearity Fat%, Weight, and Fat*Weight are to be centered. But in addition to those variables even Activity has been centered which even don't suffers from multicollinearity.

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[Reply](#)

Jim Frost says

April 5, 2020 at 7:07 pm

Hi Maheswar,

Centering the variables only reduces multicollinearity for variables that are in polynomial terms and interaction terms. If your two independent variables are included in an interaction term, you should center both of them. If they're not in an interaction term, centering will not help.

Technically, you only need to center variables with high VIFs. However, typically, if you center one variable, you should center all the continuous IVs because it affects how you interpret the constant. When you don't center IVs, the constant represents the predicted value when all IVs equal zero. However, if you center all the IVs, the constant represents the predicted value when all the IVs are at their means. If you

center only some of the IVs but not others, there's no simple interpretation for the constant. However, [I always urge caution when interpreting the constant.](#)

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[Reply](#)

gaurav sharma says

[March 18, 2020 at 4:14 am](#)

hey I am new to R, how you produced that summary containing vif and all those details. I used this code but my summary is not good and informative as yours. Neither it contains vif. Please tell you to get summary like yours. If you are using any additional package then I would like to know

``

```
linear_regressor = lm(femoral_neck ~ fat_perc + weight_kg + activity + (fat_perc *  
weight_kg), data = df)  
summary(linear_regressor)
```

``

output -

``

Call:

```
lm(formula = femoral_neck ~ fat_perc + weight_kg + activity +  
(fat_perc * weight_kg), data = df)
```

Residuals:

Min 1Q Median 3Q Max

-0.178453 -0.042754 -0.006129 0.033937 0.186795

Coefficients:

Estimate Std. Error t value Pr(> |t|)

(Intercept) 1.549e-01 1.317e-01 1.176 0.24274

fat_perc 5.571e-03 4.087e-03 1.363 0.17632

weight_kg 1.447e-02 2.852e-03 5.073 2.19e-06 ***
activity 2.238e-05 7.276e-06 3.075 0.00281 **
fat_perc:weight_kg -2.142e-04 7.394e-05 -2.898 0.00476 **

—

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07051 on 87 degrees of freedom

Multiple R-squared: 0.5623, Adjusted R-squared: 0.5422

F-statistic: 27.95 on 4 and 87 DF, p-value: 6.242e-15

“`

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[Reply](#)

Akshay Ghalsasi says

February 27, 2020 at 5:52 pm

Hi Jim,

Thanks for the excellent and intuitive explanations. I have a question regarding doing PCA to get rid of multicollinearity. Is it always wise to do a PCA before linear regression if you have high multicollinearity? If you keep all your PCA components will you get a better fit?

I recently tried PCA for Naive Bayes classification since Naive Bayes also has no multicollinearity assumption. The Naive Bayes of the PCA transformed features was worse than the non PCA transformed features, counter to my intuition. Do you have any intuition of why this could be the case?

Thanks,

Akshay

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[Reply](#)

Bianca says

February 15, 2020 at 10:03 am

Hi Jim,

Thank you so much for this article! It explained things a lot better than I've seen it written before so I will definitely be purchasing your book!

I am currently trying to create a model which predicts sales using past weather data. My variables include the daily rainfall, max temperature and min temperature and then I have calculated the sum of rainfall for the month and week prior to the sale, as well as the average max and min temperature for the month and week prior to the sale. I also have interaction factors between the max temperature and the rainfall.

Centering the variables reduced my sum of rainfall variables to an acceptable level – however the average max and min temperature for the week and month prior to the sale still have very high VIF. I guess this is because they are based on the mean anyway. Do you know if there is something I can do in this case? Am I missing something?

Thank you!

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[Reply](#)

Cat Alves says

February 6, 2020 at 5:16 am

Hey, Jim, I need some help with my master's dissertation.

I am using a simple regression, often used in literature that establishes a relationship between $\text{Price} = B_1 + B_2 \text{Equity} + \text{Net Result}$.

However, in reviewing the results I have the following problem:

-> If we analyse separately the equity variable(X) in relation to price(Y) it is significant for 5%, however in the model with net result it is no longer significant.

-> In the following year separately or together the equity variable is significant.

I am afraid that in the defence of the dissertation the question will be raised because in a model (year 2017) equity is not significant and in the following year it is.

Can you help me?

p.s: the correlation between variables and own capital is very high but close for the two years 0.8

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[Reply](#)

Jim Frost says

February 8, 2020 at 3:42 pm

Hi Cat,

First, just a minor point of terminology. Simple regression refers to regression models with a single independent variable. Because your model has more than one IV, we call it multiple regression. It's the same underlying analysis but you just have more than one IV!

Because your IVs are correlated, it's not surprising that the significance changes when you exclude one. That's known as [omitted variable bias](#). Given the high correlation, you should check the VIFs to check for severe multicollinearity. Multicollinearity can reduce the statistical power of the analysis, and might account for the insignificant result. This post talks about VIFs.

Are you fitting separate models for each year? If so, try including year as an IV and fit the model for all years and see if that helps.

Best of luck with your dissertation!

Loading...

[Reply](#)

Ravi Prasad says

February 5, 2020 at 3:01 am

Hi Jim,

Does the assumption 'multi-collinearity doesn't impact the predictive power of the model even' hold even for data sets having severe multi-collinearity among variables.

The way I see it is – Multi-collinearity certainly increases variance of the model. Before using the model built on variables having severe multi-collinearity, we should make sure the model is built on decent amount of data. Otherwise, even small changes in input data may make the model go awry.

Would like to know your views on this.

Thanks.

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[Reply](#)

Jim Frost says

February 5, 2020 at 11:32 am

Hi Ravi,

First, to clarify, that's not an assumption of the analysis, it's just how things work out. Multicollinearity (no dash) increases the variance of the coefficient estimates but it does ***not*** increase the variance of the entire model, which is why it doesn't affect the goodness-of-fit statistics.

However, your other point is valid. It is more difficult fitting the correct model in the presence of multicollinearity because the coefficient estimates and p-values are unstable. Small changes in the model can affect the coefficients and their significance greatly. It's harder to know which model is correct! A larger sample size can help but the problem won't go away completely. I talk about this aspect of multicollinearity in my post about [stepwise and best subsets regression](#) because it affects how well those procedures can select the correct model.

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[Reply](#)

Shreya Gupta says

November 23, 2019 at 11:24 pm

Hi Jim,

I checked my residual vs fitted plot for the model with all the predictor variables and the plot is a parallel line to y-axis. What does a plot like that represent. Does it represent any issues with model. Could you please share some insights on this.

Thanks

Loading...

[Reply](#)

Jim Frost says

November 24, 2019 at 2:54 am

Hi Shreya,

There's no obvious problem from your description. Is the spread of values relatively consistent across the full range of fitted values? Read this post about [residual plots](#) for more information and graphs about what they should look like.

Loading...

[Reply](#)

Shreya Gupta says

November 20, 2019 at 10:49 pm

Thanks a lot Jim !

Loading...

[Reply](#)

Shreya Gupta says

November 19, 2019 at 1:27 am

Hi Jim,

Thanks again for an in-depth explanation. Yes, by inverted sign i mean the sign of coefficient is negative (opposite) in both the models but when i plot the graph between X2 and target variable , it shows a positive linear relationship.

My question is if i use the second model with X3 removed, does inverted sign of X2 in the final regression equation to predict target cause any problem? Can i get accurate value of target with second model?

What should be done in that case?

This is the VIF after i removed X3 from base model.

Variance inflation factor for X1: 34.47

Variance inflation factor for X2: 2.66

Variance inflation factor for x4: 70.09

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[Reply](#)

Jim Frost says

November 19, 2019 at 11:10 am

Hi Shreya,

There's multiple considerations. So, let's go through them.

Negative coefficient: To start, let's refer to positive and negative coefficients, which is based on their sign. "Inverted coefficient" is non-standard terminology and using it will just confuse people. There's nothing inherently wrong with negative coefficients in general. What you need to do is determine whether the signs and magnitudes of the coefficients are consistent with theory. I write about this in my post about [choosing the correct model](#).

Graph vs. Negative coefficient: What you're seeing is a form of omitted variable bias. When you fit the model, the model accounts, or controls for, all other variables before estimating the coefficient for X_2 . However, when you graph just X_2 by Y , those other variables are uncontrolled. When you exclude important variables and you have a triangle of correlation between the excluded predictors, included predictors, and Y , you get omitted variable bias. Usually you talk about this bias in terms of regression coefficients. But, in this case, you're seeing it in your graph. Given the multicollinearity, it's not surprising you're experiencing it. For more information, read my post about [omitted variable bias](#).

Multicollinearity and coefficient signs and significance: In general, multicollinearity can cause coefficients to have the opposite sign than what they should have and tend to reduce significance. The fact that your three variables are still statistically significant despite the multicollinearity is notable. Even after removing X_3 , your model has extreme multicollinearity, as indicated by VIFs greater than 5. X_2 is not severely affected by multicollinearity. That's interesting because it has the unexpectedly negative coefficient sign. I can't tell you if X_2 's coefficient sign should be negative or not, but it's not multicollinearity that's causing it.

Multicollinearity and predictions: Yes, to make predictions, the model will use all the coefficients, including the negative X_2 coefficient. As I mention in this post, multicollinearity can flip coefficient signs and obscure statistical significance, but it all works out OK in terms of predictions. However, multicollinearity doesn't severely affect X_2 , as indicated by its low VIF. So, that coefficient is something you should look into.

Here's your to do list:

Check those [residual plots](#) for obvious problems.

Figure out whether theory suggests you'd expect a negative coefficient for X_2 .

Maybe it's not even a problem?

Consider using LASSO or Ridge regression because they can handle multicollinearity.

Because you're heavily using regression analysis, consider getting [my ebook about regression analysis](#). I think it'll be helpful!

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[Reply](#)

Shreya Gupta says

[November 18, 2019 at 11:08 pm](#)

Hi Jim,

Thanks for your reply. I removed the predictor with the highest VIF which is 256.46(X3).

Variance inflation factor for X1: 43.01

Variance inflation factor for X2: 2.66

Variance inflation factor for X3: 256.46

Variance inflation factor for X4: 140.84

Initially the adjusted r squared value was 0.901. After removing this predictor(X3) it becomes 0.898. Same is the case with RMSE, the difference is very little. In both the models, coefficient of predictor variable(X2) is inverted. Out of these two models, I don't understand which one is better, the one which suffers from multicollinearity(base model) or the one in which i have removed the predictor with the highest VIF.

Which model should be considered, though both models have negative coefficient of X2 when it should be positive. Does inverted sign of X2 create a problem in calculating Y(target variable). Also , its a very small data set , only 82 rows.

Thanks

Shreya

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[Reply](#)

Jim Frost says

November 18, 2019 at 11:48 pm

Hi Shreya,

Thanks for the extra information. Yes, that is severe multicollinearity. You need to also check the VIFs after you remove X3 to see if that resolved the problem. Removing the predictor with the highest VIF won't necessarily remove all the multicollinearity.

X2 actually isn't affected by multicollinearity. Any unusualness you're seeing with that variable isn't due to multicollinearity. When you say the coefficient is inverted, do you mean that it has the opposite sign that you'd expect theoretically? If that's the case, it's not due to multicollinearity, but there are other potential reasons. The "inverted" coefficient is only a problem if it contradicts theory. If so, you need to figure out what is happening. If it's consistent with theory, no problem! Did you check the residual plots for any obvious problems?

In terms of explanatory power, the model without X3 loses very little predictive power. The decrease in adjusted R-squared is essentially zero. It is hard to say definitively which model is better. In fact, it's possible that even the smaller model might still have too much multicollinearity. So, perhaps neither is the best. And, as I write in the post, if you're mainly interested in predicting Y, multicollinearity isn't necessarily a problem.

In general, I'd favor the smaller model because X3 was not significant to begin with and it was causing multicollinearity. Removing it has virtually no impact on the goodness-of-fit too. So, I'd lean in that direction. When you can fit a simpler model that explains the outcome virtually as well, it's often the better model. Simplicity is a good goal. But, I don't know what degree of multicollinearity the smaller model has nor fully understand what you mean by "inverted" coefficient and, therefore, don't know if that is problematic.

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[Reply](#)

Shreya Gupta says

November 18, 2019 at 6:23 am

Hello Jim.

Thanks for a great explanation.

I have built one OLS regression model with four features. It suffers from multicollinearity and one feature's p-value is greater than 0.05. When I remove the feature which is insignificant, I get another model where all the features are significant. But the adjusted R-squared value is higher for the first model and RMSE (root mean square error) is also lower for that model.

How do I now decide which model is better, the one with multicollinearity or the one without.

Thanks

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[Reply](#)

Jim Frost says

November 18, 2019 at 3:52 pm

Hi Shreya,

This is a tough issue. Multicollinearity can make things murky! It's entirely possible that the variable with the insignificant p-value should actually be significant. A classic hallmark of multicollinearity is that it makes variables appear to be insignificant even when they're not. However, the model fit statistics, such as adjusted R-squared and RMSE, are not affected by multicollinearity. That *might* explain why removing a variable that appears to be insignificant causes a large reduction in the goodness-of-fit statistics.

There's also a number of specifics that matter. How much multicollinearity do you have? Look at the VIFs with all the variables and then after you remove the one. How much do the goodness-of-fit statistics change? If it's a relatively small amount, then it's probably not a meaningful difference. If it's larger, it might mean

something, as I discuss above. If you're using the model to make predictions and don't need to understand the specific role of each variable, you probably want to leave them all in.

Unfortunately, in OLS, the variety of solutions for multicollinearity all have some type of drawback and it can be difficult to settle on the best model. Consider using LASSO or Ridge regression. Unfortunately, I don't as yet have a blog post about using them. However, I describe them in my post about [choosing the correct type of regression analysis](#). Those analyses can handle multicollinearity. You can include all your variables and obtain relatively precise estimates with the tradeoff being the introduction of a small amount of bias.

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[Reply](#)

Talha Haroon Khan says

[November 15, 2019 at 12:38 am](#)

Dependent variable is a utility function which will be obtained from these coefficients. From that utility function traffic model will be assessed.

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[Reply](#)

Ryan says

[November 14, 2019 at 4:36 pm](#)

Multicollinearity does cause inflated variance in the model which makes it unstable. Although it does not affect a model's ability to make predictions, if we introduce new data from a new sample, would it still make accurate predictions?

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[Reply](#)

Jim Frost says

November 14, 2019 at 4:42 pm

Hi Ryan, that's correct. Multicollinearity doesn't affect the model's ability to make predictions. If a model with and without multicollinearity have the same prediction precision, they should make equally good predictions. Use prediction intervals to assess that type of precision.

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[Reply](#)

Talha Haroon Khan says

November 12, 2019 at 12:11 am

Thanks a lot Sir!

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[Reply](#)

Talha Haroon Khan says

November 10, 2019 at 5:33 am

Respected Sir,

I am performing Regression Analysis on a Traffic Model.

I have 5 independent variables, i.e. Travel Time (TT), Travel Cost (TC), age, education and income level.

First two are explanatory variables and last three are just used as controllers.

The beta value for TT is opposite to the real scenario (both its sign and value), similarly,

its p value is greater than 0.05 and t less than 1.96.

How to adjust this beta for TT so that all these checks are okay?

Thanks in anticipation.

Regards,

Engr. Talha Haroon Khan

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[Reply](#)

Jim Frost says

November 10, 2019 at 6:50 pm

Hi Talha,

What's the dependent variable? You didn't say.

There's a variety of reasons for why you can obtain unexpected coefficient signs and p-values. To figure out the specific reason for your data will take some investigation. As I discuss in this post, multicollinearity can cause both unexpected signs and p-values. You might also be [misspecifying the model](#), [overfitting](#), [not fitting curvature](#), and so on.

I'd start by [checking the residual plots](#) and the [least squares assumptions](#). They can provide clues.

I have an entire chapter in [my ebook about regression analysis](#) dedicated to identifying and solving problems with your model. You should consider getting it.

Best of luck with your analysis!

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[Reply](#)

Bharat says

October 4, 2019 at 2:18 am

is it possible that the differences between groups are due to the different sample sizes ?

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[Reply](#)

Jim Frost says

October 5, 2019 at 4:18 pm

Hi Bharat,

As you increase the sample size, you'd expect the range of the data displayed in the graph to increase. However, for this example, this differences in sample size don't account for the observed differences.

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[Reply](#)

Brenda Barr says

September 18, 2019 at 10:22 pm

Hi Mr. Frost, I am a PhD candidate who is now working on the assumptions for multiple regression analysis. I have purchased your book and have read this article to hopefully help me with testing multicollinearity in the data. I have a problem that I hope you can at least help me shed light on. I chose to conduct a multiple regression analysis for my study in which I have 6 independent variables and one dependent variable. In testing the assumption of multicollinearity, the following are the numbers for Variance and for VIF. My concern are the VIF statistics for Avoidance, Distraction and Social Diversion Coping which appear to be very high. I'm lost on how to proceed.

Independent variables Tolerance VIF

Gender .935 1.070

Task Coping .434 2.304

Emotion Coping .717 1.394

Avoidance Coping .018 56.095

Distraction Coping .049 20.233

Social Diversion .050 19.904

Thanks for any insight you can offer.

Brenda Barr

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[Reply](#)

Leila says

[September 3, 2019 at 1:21 am](#)

Hi Jim,

This article is such an easy read for those of us who are statistically- challenged. Could I please ask for some advice?

I have a model with 4 IV's and one DV. I'm looking at the model with two different groups (i.e., I'm comparing the fit of the model for the 2 groups). The second group has much fewer participants (50) than the first (94).

I've found high correlations between the IV's for the second group (.65 to .82) but not for the first group. The VIF's, however, are between 2 and 4.3 for the second group. I'm unsure whether to interpret this as high multicollinearity? If so, would it be reasonable for me to simply make a composite score by adding the four IV's and seeing how this score correlates with the DV?

Also, is it possible that the differences between groups are due to the different sample sizes (94 for group 1 and 50 for group 2)?

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[Reply](#)

Miss Nadine Osei says

August 30, 2019 at 1:25 am

Hello Jim, thank you for the clarity. but i have question, it may seem damn but I want clarity on it please because I do not know which regression method to use.

When conducting my Primary research, I asked three different questions (based on my item of measure) in relation to dependent variable (purchase intention). now that i want to run a regression test, do you recommend i it three times since there are three questions for the dependent variable, or there is a way to merge.

Thanks

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[Reply](#)

Jim Frost says

August 30, 2019 at 2:47 pm

Hi, there are several different ways to answer this question.

One option is to do as you mention, create three models, one for each dependent variable.

You can linearly combine the three DVs (e.g., sum or average), if that makes sense for your research question. Then fit one model for that new DV. Going this route depends on using your subject-area knowledge and determining that a summed or average DV is somehow better than separate DVs.

You can also use a multivariate approach. For example, I write about the [benefits of using multivariate ANOVA \(MANOVA\)](#) when you have correlated DVs. You can extend this approach to including covariates if needed.

There might well be other options but those are ones that come to mind. To figure out the best option, you'll need to use your subject-area knowledge and research goals. If possible, see how other researchers have handled research in your area.

I hope this helps!

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[Reply](#)

[suvichar in hindi](#) says

August 21, 2019 at 6:57 am

Thanks a lot for sharing.I discovered a great deal of intriguing data here. A great post, extremely grateful and confident that you will compose a lot more posts like this one.

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[Reply](#)

Jim Frost says

August 21, 2019 at 11:51 am

Hi Suvichar,

I'm glad you found my website to be helpful! I'll definitely be writing a lot more! 😊

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[Reply](#)

Jerry says

August 12, 2019 at 9:41 am

Thank you so much. I am grateful sir.

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[Reply](#)

Engr.Waleed Khan says
[July 29, 2019 at 11:32 am](#)

Thank you very much. You explained it in very clear and easy words.

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[Reply](#)

Jerry says
[July 24, 2019 at 8:12 am](#)

Thank you sir for your wonderful contribution to Statistics.

My name is Jerry, I am working on Multicollinearity using Ridge Regression and R package for the analysis.

I am using the Longley Built in economic data already installed in R.

I detected severe presence of Multicollinearity by the help of VIF which were all greater than 10 (Rule of thumb).

Thereafter, I tried to use the ridge regression technique to reduce and to solve the Multicollinearity problem found.

Now, as learnt, after solving the problem of Multicollinearity in a regression model, the ridge regression coefficients are expected to be reduced and less than the OLS coefficients.

So, please I would like to know because, my ridge regression coefficients are only less than my OLS when observing the absolute coefficient values of the OLS and the Ridge Regression. So sir, is it right to make absolute comparison of our coefficients?

Thanks for your educative response.

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[Reply](#)

Jim Frost says

July 31, 2019 at 11:35 am

Hi Jerry, sorry for the delay in replying but sometimes life gets busy!

Using Ridge Regression is a great choice for multicollinearity. However, it doesn't allow you to compare the absolute values of your coefficients—at least not in most cases. For more information on what you can do, read my post about [identifying the most important variables in a regression model](#).

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[Reply](#)

ali says

July 13, 2019 at 6:37 am

I have one questions .. if i have mediating variables am i should included at the test of multicollinearity . or only independent variab;e

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[Reply](#)

Jim Frost says

July 15, 2019 at 11:07 am

Hi Ali,

In this post, I show how it's important to assess multicollinearity when you have interaction effects (a.k.a. mediating variables). So, yes, definitely check for multicollinearity for that type of variable. I can guarantee you that your model will almost definitely have high multicollinearity when it contains interaction effects or polynomials.

Fortunately, there's a simple solution—standardized your continuous IVs. I show an example of that in this post and also write about it in a post about [standardizing your variables](#).

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[Reply](#)

Raghad says

May 27, 2019 at 6:09 pm

hi sir

thanks for the important and valuable information.

I have a question please:

I have a time series data with 25 observations. the original model has a binary dependent variable and six independent variables (not binary). There is a high collinearity between variables, so I used the PCA technic to solve the problem.

Now I have a problem at the final step in the model, which is: after I got the PCA matrix and I have to run the new regression using pc1 , pc2 , and pc3 that represent 88% of the variables. shall I run the model just with pc's and deduct the othet three independent variable? because when I regret the model with 3 pc's and 3 original variables I still have multicollinearity problem.

with my respect

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[Reply](#)

philoinme says

April 28, 2019 at 4:20 am

There is this point mentioned in the third bullet under Do I have to fix multicollinearity?
– “If your primary goal is to make predictions, and you don’t need to understand the role of each independent variable, you don’t need to reduce severe multicollinearity.”

I am afraid I might be a bit off-topic but I want to know it from a data science perspective.

Will this be a disadvantage when predictions are my main goal and I don’t interpret coefficients properly or care about the statistical significance of different independent variables?

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[Reply](#)

Anastasia says

March 24, 2019 at 11:30 pm

Hi Jim!

Thank you for this post, it is definitely very helpful. I see a few people asked about detecting collinearity for categorical variables, and the suggested solution is the chi-square test. My biggest problem is that when I calculated VIFs for my linear mixed-effects model with up to 4-way interactions, some interactions of categorical variables correspond to ginormous VIFs. The interaction term seems to be collinear with one of the main effect terms included into this interaction. Those are essential interactions for my hypothesis, so I’m not sure what to do about it (sort of gave up on this for now and switched to random forests...). Am I right evaluating the collinearity in my model using the VIF? Or can/should I use the chi-square test for that?

Thanks a lot,
Anastasia

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[Reply](#)

Jim Frost says

March 24, 2019 at 11:48 pm

Hi Anastasia,

Because you're referring to VIFs, I'll assume you have continuous interactions. (VIFs are calculated only for continuous variables. Chi-squared is only for categorical variables.) Whenever you include interaction terms, you'll produce high VIFs because the main effects that comprise the interaction terms are, of course, highly correlated with the interaction terms. Fortunately, there is a very simple solution for that, which I illustrate in this blog post. Simply center all of your continuous independent variables and fit the model using the centered variables. That should cut down those VIFs by quite a bit. Often centering will get the VIFs down to manageable levels. So, reread that section of this post!

I hardly ever see models with three-way interactions, much less four-way interactions. Be sure that they're actually improving the model. If the improvement is small, consider removing them. Higher-ordered interaction terms like those are very difficult to interpret. Of course, your model might be much better with those terms in it. But, at least check because it is fairly unusual.

I hope this helps!

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[Reply](#)

JP says

March 11, 2019 at 11:54 am

Hi Jim, Thanks for the great article, I am working on multiple regression model where my VIF values are less than 2 for all the variables. From the model one variable changes sign of the coefficient (from the theory) even when the VIF value is less than 2 , I

checked that one variable with DV it is showing the right coefficient sign and it is not statistically significant on its own . Should i look into interaction with that variable.

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[Reply](#)

Jim Frost says

March 11, 2019 at 3:16 pm

Hi JP,

With VIFs that small, it's unlikely that multicollinearity is flipping the signs of the coefficients.

Several possibilities come to mind that can flip signs. The two most common are issues that I've written about: [Omitted variable bias](#) and [incorrectly modeling curvature that is present](#). For a quick check on the curvature issue, check the [residual plots](#)! And, yes, it's possible that not including an interaction term that should be in the model can cause this problem. All of these issues are forms of model specification error (i.e., your model is incorrect and giving you biased results). I'd check those things first.

I'm assuming that the independent variable that has the incorrect sign is statistically significant?

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[Reply](#)

Zara says

March 7, 2019 at 3:45 pm

Hello! Im doing a regression analysis where I have 2 measurements of the same concept. Since the measurements are very correlated I was thinking of creating a

composite variable by averaging their z-scores. Is this a good step and would the interpretation of the regression change if I'm using standardized scores?

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Reply

Jim Frost says

March 7, 2019 at 4:15 pm

Hi Zara,

The approach you describe is definitely an acceptable one. First, I'd fit the model with both measures and check the VIFs as I describe in this post. Be sure that's actually a problem. The two variables have to be very highly correlated to cause problems.

If it is a problem, I'd also compare the goodness-of-fit for models with both measurements versus models with just one of the measurements. If they are very highly correlated, you might not lose much explanatory power by simply removing one because they're supply highly redundant information to begin with. If the change in goodness-of-fit is small, you can consider including only one measure. This approach gives you the ability to estimate the effect of one measurement.

However, if you do decide to combine the measurements as you describe, it does change the interpretation. You're standardizing the measurements. For standardized data, the coefficients represent the mean change in the dependent variable given a change of one standard deviation in the independent variable. That's still a fairly intuitive interpretation. However, you'd be averaging the z-scores, so it's not quite that straightforward. Instead, the interpretation will be the mean change in the DV given the average change of one standard deviations across those two measurements. You'll lose the ability to link the change in a specific IV to the DV. However, it's possible you'll gain more explanatory power. Before settling on this approach, I'd check to be sure that it actually improves the fit of the model.

It's possibly a good approach depending on whether the VIFs are problematic, the loss of explanatory power by just using one measurement, and the gains in explanatory power by averaging the two.

Best of luck with your analysis!

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[Reply](#)

Rui Fang says

March 6, 2019 at 11:45 pm

Hi Jim,

Can I use VIF to test for multicollinearity between categorical independent variables?

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[Reply](#)

Jim Frost says

March 7, 2019 at 9:45 am

Hi Rui,

VIFs only work for continuous IVs. I think you'd need to use chi-squared test of independence with categorical variables. It is harder to determine with categorical variables.

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[Reply](#)

Nhat T. Tran says

February 28, 2019 at 7:24 am

Hi Jim,

Thank you for your reply. If it would not take up too much of your time, I would like to ask you more one question.

As you mentioned in the example, using standardized variables does not influence the predictions, so my primary goal is to investigate relationships of independent variables with a dependent variable.

In your example, after using the centered independent variables, the sign for %Fat has changed from positive to negative. Using subject-area knowledge, if %Fat has a negative relationship with the bone mineral density of the femoral neck, we may pick the standardized solution. On the other hand, if %Fat has a positive relationship with the bone mineral density of the femoral neck, we may choose the uncoded model.

I wonder whether the above solutions are appropriate or not. Therefore, I would appreciate any assistance or comments you could give me. Thank you again for your time and consideration.

Best regards,
Nhat Tran.

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[Reply](#)

Jim Frost says

February 28, 2019 at 11:27 am

Hi Nhat,

Unfortunately, because you're interested in understanding the relationships between the variables, this is not a case where you can choose between these two models based on theory. A model with excessive multicollinearity is one that has a problem that specifically obscures the relationships between the variables. Consequently, you don't want to choose the model with multicollinearity because it

happens to agree with your theoretical notions. That's probably just a chance agreement!

What you really need to do is find a method that both resolves the multicollinearity and estimates relationships that match theory (or at least you come up with an explanation for why it does not). If I wasn't able to use centering to reduce the multicollinearity, I'd probably need to use something like Ridge or LASSO regression to accomplish this task.

If the model with acceptable multicollinearity produces estimates that don't match theory, consider the possibility that you're specifying the incorrect model. That would be the next issue I'd look into. But, don't go with the model that has problematic multicollinearity just because it happens to agree with your expectations.

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[Reply](#)

Nhat T.Tran says

[February 23, 2019 at 7:05 am](#)

Hi Jim,

Thank you so much for creating a great blog for statistics.

In the ANOVA result of your example, you used the adjusted sums of squares (Adj SS). I also run the same test for your data with Minitab using sequential sums of squares (Seq SS). However, two results are different as follows.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	0.555785	0.138946	27.95	0.000
%Fat	1	0.009240	0.009240	1.86	0.176
Weight kg	1	0.127942	0.127942	25.73	0.000
Activity	1	0.047027	0.047027	9.46	0.003
%Fat*Weight kg	1	0.041745	0.041745	8.40	0.005

Error 87 0.432557 0.004972

Total 91 0.988342

Analysis of Variance

Source	DF	Seq SS	Seq MS	F-Value	P-Value
Regression	4	0.55578	0.138946	27.95	0.000
%Fat	1	0.20514	0.205137	41.26	0.000
Weight kg	1	0.24506	0.245059	49.29	0.000
Activity	1	0.06384	0.063843	12.84	0.001
%Fat*Weight kg	1	0.04175	0.041745	8.40	0.005
Error	87	0.43256	0.004972		
Total	91	0.98834			

The ANOVA results using Seq SS shows a statistically significant effect of %Fat on the Femoral Neck (P value = 0.000), while The ANOVA results using Adj SS indicates there is not a statistically significant effect of %Fat on the Femoral Neck (P value = 0.176).

Therefore, I wonder that when will we use “sequential sums of squares” and when will we use “adjusted sums of squares”.

If you could answer all two of my questions, I would be most grateful.

Best regards,

Nhat Tran.

Loading...

[Reply](#)

Jim Frost says

February 24, 2019 at 6:56 pm

Hi Nhat,

Quick definitions first.

Adjusted sums of squares: Calculates the reduction in the error sum of squares for each variable based on a model that already includes all of the other variables in the model. The procedure adds each IV to a model that already contains all of the other IVs, and determines how much variance it accounts for.

Sequential sums of squares: Calculates this reduction by entering the variables in the specific order that they are listed. The procedure enters the first variable first, then the second variable, third, and so on.

The analysis uses these sums of squares to calculate F-values and t-values, which in turn determines the p-values. So, it's not surprising that changing the sums of squares affects the p-values.

The standard choice you almost always want to use is the Adjusted Sums of Squares (Type III). Using this method, the model can determine the unique portion of variance that each variable explains because it calculates the variance reduction for each variable when it is entered last. This type of SS is used for at least 99% of the regression models! Basically, use this type unless you know of very strong reasons to the sequential sums of squares.

I don't have a strong case for using sequential sums of squares. You'd need really strong theoretical reasons for why the variables need to be entered into the model in a specific order. This option is almost never used that I'm aware of.

I hope this helps!

Loading...

[Reply](#)

[akshay thakar](#) says

[February 2, 2019 at 8:11 am](#)

Thanks for such a quick response !! The chi square test for Independence can involve only 2 categorical variables at a time, so should I take multiple pairs one by one to check for multi collinearity...?? Or is there any way to do the chi square test for multiple variables..??

Loading...

[Reply](#)

Jim Frost says

February 4, 2019 at 2:57 pm

Hi again Akshay,

Yes! You can definitely use additional variables. In chi-squared, they're referred to as layer variables. Although, it can get a bit unwieldy when you have many, I'd try that approach. I can help show if you potentially have a problem and where to look for it!

Loading...

[Reply](#)

akshay thakar says

February 1, 2019 at 10:36 am

Hi Jim,

Your blog is amazing !!

I am trying to run a regression analysis however I am facing the issue of multi-collinearity between categorical variables , are there any tests to identify multi-collinearity between categorical variables ??

Loading...

[Reply](#)

Jim Frost says

February 1, 2019 at 11:53 am

Hi Akshay, thanks for the kind words!

The chi-squared test of independence would be a good way to detect correlations between categorical variables. I cover this method in this blog post: [Chi-squared](#)

Test of Independence and an Example.

Loading...

[Reply](#)

Lola says

January 29, 2019 at 1:47 pm

Thank you! Thank you!

Loading...

[Reply](#)

Lola says

January 28, 2019 at 5:26 am

Thanks a bunch Jim!

Is that (pairwise correlation) the same as producing a correlation matrix of all the independent variables from the regression model?

I'd also like to know if and how you take classes . I'd really want to hone my statistical 'skills'.

Loading...

[Reply](#)

Jim Frost says

January 28, 2019 at 9:30 am

Hi Lola, you're very welcome!

You could certainly do a matrix of all the variables as you describe. However, that might provide more correlations than you need.

Suppose X_1 is the IV with the high VIF and that the others have low VIFs. You'd really just need the correlations of X_1 with X_2 , X_3 , X_4 , and so on.

If you did a matrix, you get all the correlations with say X_2 and X_3 , X_2 and X_4 , etc. You might not need those. It doesn't hurt to obtain extra correlations, it's just more numbers to sort through!

Just keep in mind that only the terms in the model with high VIFs are actually affected by multicollinearity. You can have some terms with high VIFs and others with low VIFs. Multicollinearity does not affect the variables with low VIFs,, and you don't need to worry about those.

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[Reply](#)

Lola says

January 26, 2019 at 11:08 am

Hi Jim, this page is absolutely brilliant. Thanks for this initiative.

However, I have a question on something a bit unclear regarding the VIF interpretation. In the first regression model above, it can be seen that 'Fat', 'Weight' and 'Fat*Weight' have "worrisome" VIF values., which depict multicollinearity. And as explained above, multicollinearity involves 2 or more independent variables.

the question is – For each, Independent variable with worrisome VIF values, how does one determine which of the other IVs, it is highly correlated with?

Loading...

[Reply](#)

Jim Frost says

January 27, 2019 at 11:03 pm

Hi Lola,

That is a great question! The VIF for a specific term in model shows the total multicollinearity across all of the other terms in the model. So, you're right, seeing a high VIF indicates there is a problem but it doesn't tell which variable(s) are the primary culprits. At that point, you should probably calculate the pairwise correlations between the independent variable in question and the other IVs. The variables with the highest correlations would be the primary offenders.

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[Reply](#)

Ioakim Boutakidis says

January 13, 2019 at 1:20 pm

Greetings Jim...just wanted to drop a quick note and say that you have laid out some excellent content here. I landed on your cite after a student of mine came across it looking for info on multicollinearity, and so I felt I had to check it out to make sure she was getting legitimate information. I have to say I am very impressed. It looks like you are helping a lot of people do better research, and that's something you should be very proud of.

Loading...

[Reply](#)

Jim Frost says

January 13, 2019 at 9:25 pm

Hi Ioakim, thank you so much for you kind words. They mean a lot to me! I'm glad my site has been helpful for your students!

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[Reply](#)

Michal says

December 31, 2018 at 2:54 am

Very much! Thank you!

Loading...

[Reply](#)

Michal says

December 30, 2018 at 9:35 am

Could it be that two highly related independent variables ($r=0.834!!$) yield a VIF of 3.82? Of course, it makes my life easier that I don't have to deal with the multicollinearity problem, but I don't understand how this can happen....

Loading...

[Reply](#)

Jim Frost says

December 31, 2018 at 2:30 am

Hi Michal,

Yes, that might be surprising but it is accurate. In fact, for the example in this blog post, the %fat and body weight variables have a correlation of 0.83, yet the VIF for a model with only those two predictor variables is just 3.2. That's very similar to your situation. When you have only a pair of correlated predictors in your model, the correlation between them has to be very high (~ 0.9) before it starts to cause problems.

However, when you have more than two predictors, the collective predictive power between the predictors adds up more quickly. As you increase the number of predictors, each pair can have a lower correlation, but the overall strength of those relationships accumulates. VIFs work by regressing a set of predictors on another predictor. Consequently, it's easier to get higher VIFs when you have more predictors in the model. No one predictor has to "work very hard" to produce problems.

But, when you have only two predictors, the relationship between them must be very strong to produce problems!

I hope this helps!

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[Reply](#)

Ben says

November 9, 2018 at 5:27 pm

What if the sole purpose of the regression is to identify the "rank" of the contributions of the independent variables? If all the variables (including the dependent variable) are all correlated with each other, does this "drivers analysis" still hold?

Loading...

[Reply](#)

Javed Iqbal says

November 5, 2018 at 5:36 am

Q L K

11 12.2 10.1

34.6 30.2 28.2

21.9 23 24

28.2 22.3 21.3

14.7 15.7 14.3

20.2 20.8 18.4

9.7 11.5 10

22.2 25.9 24

17.3 21.5 20.3

19.5 22.4 20.5

13.6 14.4 12.2

34 29.5 29.2

35.1 26.8 25.5

10.6 12.7 10.8

18.6 19.6 19.9

22.9 25 24

27.4 25.7 23.2

16.4 18 16.2

22 18.3 19.4

27 19.7 17.2

27.1 23.7 25

15.6 21.2 20.5

13.2 23 22.1

27.3 26.3 24.3

15.4 22.6 20.8

30.6 30.5 28.9

24.4 28.6 28.1

36.1 26.7 27.9

24.8 21.7 20.7

21 18.5 17.1

10.2 13.5 11.1

20.4 13.4 11.7

14.3 15.7 15.4

This is the data 'cobb' from Hill-Griffith-Lim, Principles of Econometrics. The estimate of the Production function results in the following (with R-sq of 0.69 and overall F of 33.12 with p-value of 0.000). This is a classical case of multi collinearity. as none of the individual coefficients are significant. The sample correlation b/w log of L and log K is 0.985 and the VIFs are 35.15 for both variables. I will appreciate if you could comment on resolving the multicollinearity issue.

Variable Coefficient Std. Error t-Statistic Prob.

C -0.128673 0.546132 -0.235608 0.8153

LOG(L) 0.558992 0.816438 0.684672 0.4988

LOG(K) 0.487731 0.703872 0.692925 0.4937

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[Reply](#)

derekness says

[October 29, 2018 at 4:20 am](#)

Hi Jim,

thanks again for the useful input.

I have been playing with the model and trying to see how it responds when I do put in co-linear data in. If I put in the two strong peaks for A and B twice. It goes mad and falls over. This is good. I think it tells me that the additional data contained in the ratio of the two strong peaks is different than the data contained in the two individual peaks. This is good as it really appears to help the model work and give me great predictive powers. It gives me confidence that all is working well.

I am also working to improve the performance of the model on how it handles independent data.

We have to make calibration mixes of A and B in the lab. We can do this very accurately and have a great machine for making a super homogenous air free mixes. The model uses these mixes with varying amounts of A and B to then be able to test mixes made in

a production environment from a large metering and mixing machine. Unfortunately these do not “look” like the perfect lab made mixes and the analysis gives me a wide variations in the compositions of A and B. I think that this is not real. I therefore am now tuning the model so that it handles the production mixes better. This makes the R^2 values for the calibration model worse, but I now get much better R^2 values from the production mixes. This approach has worked well, and the model is now insensitive to whatever is different in the real world production mixes.

This has been a great learning process for me (and also a lot of fun), but I always am cautious as it is a bit of a “black box” and I have no idea what it is up to!

Oh and yes we also use PLS treatments for some analysis’s, and that can be really good on tricky materials. For this one I have to use the ILS method, but it seem pretty good. With the PLS work you have more insight to what is going on with the factors and PRESS data, the ILS software doesn’t let on how it does it!

regards,

Derek.

Loading...

[Reply](#)

Jim Frost says

[October 29, 2018 at 11:10 am](#)

Hi Derek,

Thanks for the follow-up! I always enjoy reading how different fields use these analyses. While the methods are often the same, the contexts can be so different!

It does sounds like you have a promising model. And, the super accurate machine you have explains how you can obtain a very high R^2 when you assess the lab mixes. As you found, you’d expect the real life mixtures to have a lower R^2 .

That actually reminds me of research done in the education field. When some researchers tested a new teaching method, they did so in a very controlled lab-like setting. It worked in that setting. However, when they tried it in the field, their

results were not nearly as good. They learned that because the new method had to work in the field that it had to be robust to variations you'd expect in the field. You always think of reducing the variability for experiments, but there's also the need to reflect the actual conditions. And, that sounds kind of like what you're dealing with.

It's great that it's been a fun experience for you! That's the way I see it too and what I always try to convey through my blog. That statistics can be fun by helping you discover how something works. It's the process of discovery.

Thanks for sharing!

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[Reply](#)

[Derek Ness](#) says

[October 28, 2018 at 8:18 am](#)

Jim, thank for that. We have a good grasp of the chemistry but the maths that we use we don't understand and have no feel for at all. The FTIR spectra is just a collection of peaks. We do one for component A and another for component B. We look for areas where we get a strong peak in one component and nothing in the other. We then use these peaks to put into the Inverse Least Squares (I think it is actually a MLR) treatment, on a series of mixes with varying amounts of A and B in. I do this and I get an OK model. If I then add in the ratio of the two peaks as well as the individual peaks, the model gets amazingly good, R^2 0.9999 from R^2 0.9. It also gets very good at analysing my set of independent data. I am mentally struggling with whether this is real or is the treatment cheating, and it will all fall apart when I get a new set of data to analyse.

I will probably work up both approaches and use them on the next application and see if the super one keeps looking great!

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[Reply](#)

Jim Frost says

October 28, 2018 at 4:39 pm

Hi Derek,

I don't know enough about the subject to have an opinion on whether it is "cheating" or valid. You should think through the logic of the model and determine whether it makes sense. I wouldn't say there is anything inherently "cheating" about including the ratio. But, does that make sense from a subject matter point of view. What was your rationale for including it? And, do the coefficient estimates fit your theory?

Also, consider your sample size. Just be sure that you're not [overfitting the model](#). It doesn't sound like you have too many predictors, but just something to consider.

Ultimately, yes, I think cross validation with a new dataset is the best way to evaluate a model. I think that's always true. But, even more so when you have questions like the ones you have! Sounds like you've got a great plan to address that! I'd be curious to hear how that goes if you don't mind sharing at a later point?

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[Reply](#)

Derek Ness says

October 27, 2018 at 6:39 am

Jim, I am working on FTIR spectra of mixes and we are using a ILS treatment to build an analysis model. We can select peaks to include in the model we can also use ratios of peaks. As I see it if I use two strong peaks from component A and two strong peaks from component B I have a colinearity issue as these peaks are related. Does this cause a problem?

Right now I am using a single peak from A and a single peak from B. This works OK, but if I then use a ratio of these 2 peaks as well the model looks amazing with an Rsq. of 1.

This looks suspicious to me. Is colinearity effects causing this and is it thus dangerous to use the ratio and the 2 individual peaks. The cross validation for this also looks great and it appears to analyse an independent set of data really well.

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[Reply](#)

Jim Frost says

October 27, 2018 at 5:03 pm

Hi Derek,

First, let me say that I know so little about the subject area, which will limit what I can say. Statistics should always be a mix of statistical knowledge and subject-area expertise.

If your goal is to make a good prediction, then you don't need to worry about multicollinearity. It's often surprising, but multicollinearity isn't bad when it comes to the goodness-of-fit measures, such as R-squared and S. It does affect the precision of the coefficient estimates and their p-values. But, if the coefficients/p-values aren't your main concern, multicollinearity isn't necessarily a problem.

Again, I don't know the subject area, but for physical phenomenon, it's not impossible to obtain very high R-squared values if there's very little noise/random error in the process. That's something you'll have to determine using your subject-area knowledge. Maybe research what others have done and the results they obtained. But, again, the very high R-squared might not be a problem. I did write a post about [reasons why your R-squared value might be too high](#), which you should read because it covers other potential reasons why it could be too high. But, the fact that your cross validation looks good is a great sign!

Finally, I'm also aware that analysts often use partial least squares (PLS) regression for analyzing spectra because of both a large number of predictors and multicollinearity. This form of regression is a mix between principle components analysis and least squares regression. I'm not sure if it would be helpful for your analysis, but it's an analysis to consider. Unfortunately, I don't have much firsthand experience using PLS so I don't have much insight to offer. But, if you need to consider other forms of analysis (which you might not), it's one I'd look in to.

I hope this helps. Best of luck with your analysis!

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[Reply](#)

Jerry Avura says

October 26, 2018 at 4:43 am

Hello sir, how can one calculate for VIF using R? Thanks in anticipation.

Loading...

[Reply](#)

Jim Frost says

October 26, 2018 at 10:41 am

Hi Jerry,

Unfortunately, I don't use R and don't know the answer. Sorry.

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[Reply](#)

Jerry Avura says

October 2, 2018 at 3:08 am

thank you sir. Looking forward to it

Loading...

[Reply](#)

Jerry Avura says

October 1, 2018 at 9:05 am

Thank you sir for the nice job, it was so clear and perfect. I'm Jerry, an MSc Student of Statistics. I'm currently working on Multicollinearity. Please can you throw more light on "Ridge Regression"?

Thanks in anticipation.

Loading...

[Reply](#)

Jim Frost says

October 1, 2018 at 9:20 pm

Hi Jerry, at some point I'll try to write a blog post about Ridge Regression, but I have a bit of a backlog right now! I do talk about it a bit in my post about [choosing the correct type of regression analysis](#).

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[Reply](#)

John Komlos says

September 24, 2018 at 11:13 am

that's interesting. thank you very much. I did not know that. do you have a citation for me by any chance? thanks in advance, John

Loading...

[Reply](#)

Jim Frost says

September 24, 2018 at 11:39 am

You bet! That's a generally recognized property of multicollinearity so any linear model textbook should discuss this issue. In this post, I include a reference to my preferred textbook for another issue. That's the one I'd recommend, but any good textbook will talk about this issue. I don't know of any articles offhand.

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[Reply](#)

John Komlos says

September 24, 2018 at 10:39 am

Thank you Jim, appreciate the explanation. One more question: would it be possible for the two variables to be significant in spite of multicollinearity?

Thanks.

Best regards,

John

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[Reply](#)

Jim Frost says

September 24, 2018 at 11:00 am

Hi, it's definitely possible. While multicollinearity weakens the statistical power of the analysis, it's still possible to obtain significant results—it's just more difficult. Additionally, the coefficient estimates are erratic and can swing widely depending

on which variables are in the model. While you can obtain significant results, this instability makes it more difficult for you to be confident in which specific estimates are correct.

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[Reply](#)

John Komlos says

September 23, 2018 at 10:29 pm

I wonder if two multicollinear variables can be both statistically significant. one is large and negative while the other is large and positive and both significant. i have a feeling that like magnets they repel each other. is that possible?

Loading...

[Reply](#)

Jim Frost says

September 24, 2018 at 10:31 am

Hi John,

It's certainly possible for multicollinear variables to have opposite signs like you describe. However, there is no propensity for that situation to occur. That is to say, having different signs or the same signs are equally likely and just depends on the nature of the correlations in your data. The real issue is that you can use one independent variable to predict another. It's really the absolute magnitude of the correlation coefficient that is the issue rather than the signs themselves.

It actually gets a bit more involved than that. VIFs aren't just assessing pairs of independent variables. Instead, VIF calculations regress a set of independent variables on each independent variable. It's possible that two or more independent variables collectively explain a large proportion of the variance in another

independent variable. In a VIF regression model, it's possible to have a mix of positive and negative signs!

That's probably more than you want to know! But, to your question, yes, it's possible but it's really the absolute magnitude that is the issue.

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[Reply](#)

Patrik Silva says

September 21, 2018 at 5:13 am

Hi Jim, Thank you, I got your point! You are helping me a lot!

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[Reply](#)

Patrik Silva says

September 20, 2018 at 10:23 am

Hi, Jim

I would like to know if, when you mentioned: Fat S, Weight S , Activity S and FatS * WeightS. The Fat S multiplied by Weight S (Fat S * Weight S) is calculated using the Fat S (Standardized) * Weight S (Standardized) or is the standardized of the two variable together by taking the (Fat * Weight) S.

I do not know if you got my point!

Thank you!

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[Reply](#)

Jim Frost says

September 20, 2018 at 3:33 pm

Hey Patrick, it's the first scenario that you list.

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[Reply](#)

Patrik Silva says

September 20, 2018 at 8:59 am

Dear Jim, you are making people love statistic! Every time I come here to read something, I am getting more love to "Stats". You explain statistics so easy, but so easy that I feel like I am reading/hearing a beautiful story.

Thank you Jim!

Loading...

[Reply](#)

Sinan says

August 20, 2018 at 4:59 am

Hi Jim,

It can be used methods such as backward elimination for property selection in multiple linear regression (MLR) . In these methods, features are removed from the model, just like in Multicolliniarity. The question I want to ask is: In an MLR application, which one to do first? Multicolliniarity or model selection?

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Reply

Jim Frost says

August 23, 2018 at 2:17 am

Hi Sinan,

This is a tricky situation. The problem is that multicollinearity makes it difficult for stepwise regression (which includes the backward elimination method) to fit the best model. I write about this problem in my post that [compares stepwise and best subsets regression](#). You can find it in the second half of the post where I talk about factors that influence how well it works.

However, removing multicollinearity can be difficult. But, I would try to remove the multicollinearity first.

There is another approach that you can try—LASSO regression. This method both addresses the multicollinearity and it can help choose the model. I describe in my post about [choosing the right type of regression analysis to use](#). I don't have hands-on experience with it myself, but it might be something you can look into if it sounds like it can do what you need it to do.

I hope this helps!

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Reply

Veikko says

August 8, 2018 at 6:44 am

Hi, do you have an author for your reference "Applied Linear Statistical Models"?

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Reply

Jim Frost says

August 8, 2018 at 10:36 am

Michael H. Kutner et al.

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[Reply](#)

John Velez says

April 9, 2018 at 12:31 pm

Hi Jim,

Again, thank you for such a great explanation!

My question is as follows:

Say you have two “severely” correlated IVs (X_1 and X_2) but you’re interested in examining each one individually. How would controlling (i.e., enter as a covariate) X_2 while examining X_1 influence your coefficients and p values? I would assume a loss of power but what else may occur? I’m also interested in any potential pitfalls of using this approach.

Thanks for your time!

John

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[Reply](#)

Jim Frost says

April 9, 2018 at 4:05 pm

Hi John,

There are several problems with including severely correlated IVs in your model. One, it saps the statistical power. However, it also makes the coefficients unstable. You can change the model by including or excluding variables and the coefficients can swing around wildly. This condition makes it very difficult have confidence in the true value of the coefficient. So, the lower statistical power and unstable coefficients are the major drawbacks. Basically, it's hard to tell which IVs are correlated with the DV and the nature of those relationships.

One thing I don't mention in this post, but I should add, is that you can try Ridge regression and Lasso regression, which are more advanced forms of regression analysis that are better at handling multicollinearity. I don't have much first hand experience using them for that reason but they could be worth looking into. I mention them in my post about [choosing the correct type of regression analysis](#).

I hope this helps!

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[Reply](#)

seyi says

[March 26, 2018 at 3:15 pm](#)

Hi Jim,

Just to thank you for the clear explanation on your articles.

Loading...

[Reply](#)

Jim Frost says

[March 28, 2018 at 3:02 pm](#)

Hi Seyi, you're very welcome! Thanks for taking the time to write such a nice comment! It means a lot to me!

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[Reply](#)

Filmon says

March 15, 2018 at 2:08 am

You are just amazing Mr. Jim Thank you for the wonderful and exhaustive note

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[Reply](#)

Jim Frost says

March 15, 2018 at 10:26 am

You're very welcome! Thanks for taking the time to write such a kind comment!

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[Reply](#)

Vijay says

March 8, 2018 at 5:46 am

Hello Jim Frost,

It's Awesome explanation regarding Multicollinearity.

But i have one doubt, is there any other method to detect multicollinearity except VIF?

Thanks in advance.

Regards,
Vijay

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[Reply](#)

Jim Frost says

March 8, 2018 at 10:22 am

Hi Vijay, thank you for your kind words! Thanks for writing with the great question!

VIFs really are the best way because they calculation the correlation between each independent variable with ALL of the other independent variables. You get a complete picture of the combined strength of the correlation.

You can also assess the individual simple correlations between pairs of IVs. This approach can tell you if two IVs are highly correlated, but can miss out on correlations with multiple IVs. For instance, suppose that individually, IV1 has a moderate (but not problematic) simple correlation with IV2 and IV3. However, collectively IV1 has problematic levels of correlation with both IV2 and IV3 combined. VIFs will detect the problem while simple correlation will miss it.

I hope this helps!

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[Reply](#)

Blake Wareham says

February 1, 2018 at 5:21 am

Hi, Jim! Third year MA/PhD student here, very much appreciating the time and work you've put into communicating these concepts so effectively. I've been reading through

a few of your posts for information about a three-way interaction in a regression used for my thesis, and will certainly cite your pages. I'm also wondering if it would also be possible to provide other sources in these posts that reiterate or elaborate on the concepts? I imagine at least some of it includes Dawson, Preacher, and perhaps the Aiken and West paper.

Thanks again for what you're doing here.

Best,
Blake

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[Reply](#)

Jim Frost says
[February 1, 2018 at 11:38 am](#)

Hi Blake,

Thanks for the kinds words! They mean a lot!

I cover [two-way interactions in a previous post](#). However, I find the interpretation of three-way interactions to be much more complicated. Suppose the three-way interaction $A*B*C$ is statistically significant. This interaction indicates that the effect of A on the dependent variable (Y) depends on the values of both B and C. You can also come up with similar interpretations for the effects of B and C on Y.

In my post about two-way interactions, I show how graphs are particularly useful for understanding interaction effects. For three-way interactions, you'll likely need even more graphs. If you're using categorical variables in ANOVA, you can also perform post-hoc tests to determine whether the differences between the groups formed by the interaction term are statistically significant.

Down the road, I might well write a more advanced post about interpreting three-way interactions. However, in the meantime, I hope this explanation about [interpreting three-way interaction effects](#) is helpful.

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[Reply](#)

Nivedan says

December 20, 2017 at 7:07 am

Hi Jim!

What if a categorical independent variable (converted to dummies) has 2, 3 or more category levels and are severely correlated?

Regards

Nivedan.

Loading...

[Reply](#)

Jim Frost says

December 20, 2017 at 1:33 pm

With correlated categorical, independent variables, you'll face the same problems as with any correlated independent variables. You can also use the same potential solutions—partial least squares, Lasso regression, Ridge regression, removing one of them, combining them linearly, etc.

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[Reply](#)

Avinash says

November 28, 2017 at 8:09 am

Thanks, Jim!

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[Reply](#)

Avinash says

November 27, 2017 at 8:13 pm

Hi Jim,

If one independent variable is a subset of another independent variable, then does that also cause problems? Say for ex, I'm looking at no. of people who have completed high school and also no. of people with a bachelor's degree. The former is clearly a subset of the latter. Should they be separately considered in 2 different models to study the effects?

And a second question – what about a cluster analysis? Does it cause any problem to have 2 correlated variables?

Loading...

[Reply](#)

Jim Frost says

November 28, 2017 at 2:01 am

Hi Avinash,

In your education example, yes, those variables would clearly be correlated. Individuals with a bachelor's degree must also have HS degree. And those without a HS degree cannot have a bachelor's degree. Usually, analyses will include a variable for the highest degree obtained (or worked on), which eliminates that problem.

Correlated variables can affect cluster analysis. Highly correlated variables are not sufficiently unique to identify distinct clusters. And, the aspects associated with the correlated variable will be overrepresented in the solution.

I hope this helps!

Jim

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[Reply](#)

prabuddh says

October 20, 2017 at 5:45 pm

i am still trying to figure out the solution and will share here once i get it. thank you for your response though. and great articles to understand concepts.!

Loading...

[Reply](#)

Jim Frost says

October 21, 2017 at 11:12 pm

I would love to hear what you do. Please do share! Best of luck!

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[Reply](#)

Prabuddh says

October 15, 2017 at 7:05 am

In case of marketing mix where we want to understand the impact of say radio, TV, internet and other media campaigns individually, we cannot remove or merge these variables. How to go about them? And obviously these variables highly correlated

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[Reply](#)

Jim Frost says

October 15, 2017 at 12:57 pm

Hi, first you should make sure that they're correlated to a problematic degree. Some correlation is OK. Check those VIFs! If they're less than 5, then you don't need to worry about multicollinearity. I'm not sure that I'd necessarily expect the correlation between the different types of media campaigns to be so high as to cause problems, but I'm not an expert in that area.

If it is too high, you need to figure out which solution to go with. The correct solution depends on your specific study area and the requirements of what you need to learn. It can be a difficult choice and not one that I can make based on a general information. In other words, you might need to consult with a statistician. You'd have to choose among options such as removing variables, linearly combining the measures, using principle components analysis, and partial least squares regression.

You've already ruled a couple of those out but there are still some options left. So, I'd look into those.

I hope this helps!

Jim

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Sibashis Chakraborty says

September 20, 2017 at 3:52 pm

say I am regressing Y on two variables x_1 and x_2 and x_1 and x_2 has a high correlation between them. If I run a regression of Y on x_1 and then Y on x_2 and finally Y on x_1 and x_2 I will get different values for coefficients that will affect my prediction. don't you think?

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Jim Frost says

September 22, 2017 at 9:04 pm

Hi, it depends on all of the specifics. However, if X_1 and X_2 are highly correlated, they provide very similar information in model. Consequently, using X_1 , X_2 or, X_1 and X_2 might not change the predictions all that much. As the correlation between the two increases, this becomes ever more the case.

I hope this helps!

Jim

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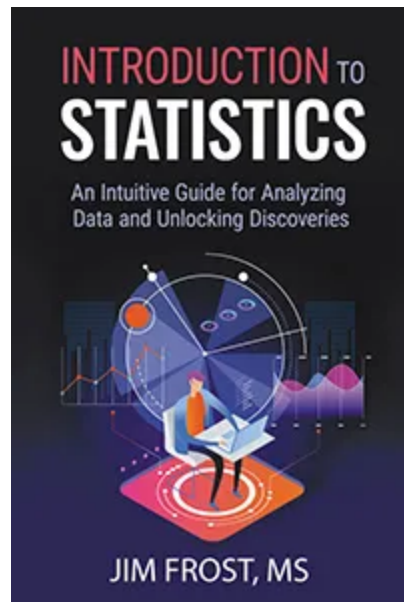
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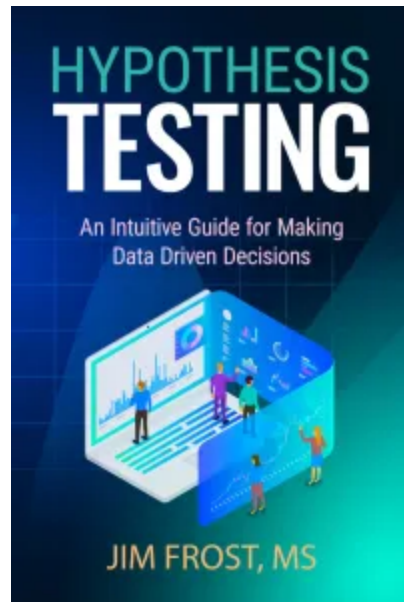
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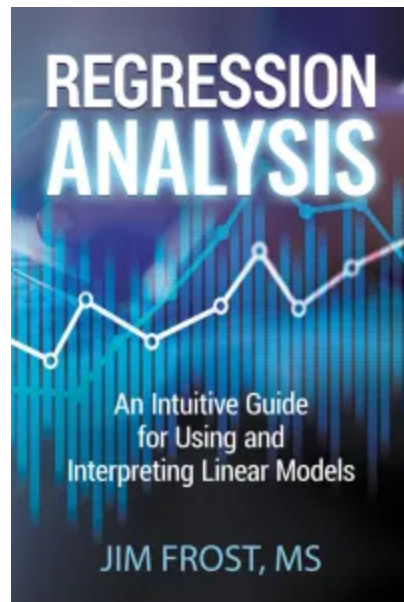
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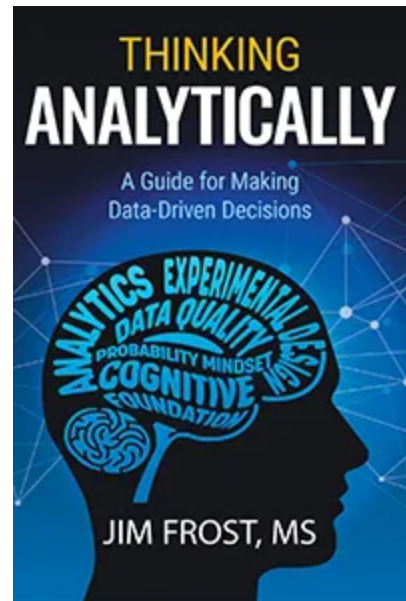
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